ABSTRACT

Warehouse fires are one of problems that threaten sustainability of the warehouse. Early prevention system for indications of warehouse fires is absolutely necessary. The extent of the warehouse to be one of the problems encountered in the warehouse condition monitoring. To overcome the problems of warehouse, designed a system of warehouse fire prediction system by adopting the Wireless Sensor Network (WSN) using multiple sensor nodes with IoT module. In this proposed system, detection and Monitoring of warehouse fires through several sensors and send to IoT cloud Depending upon the sensor values in the cloud if it is greater than the threshold values it will send to the cloud server. Continuous monitoring and uploading values to cayenne cloud can be achieved. Each sensor node has a microcontroller, IoT module, and fire and smoke (temperature, flame, gas sensor) detection sensors. A various gas sensor (gas sensor are used to detect the harmful gas, the temperature sensor and flame is LM 35 to monitor the temperature, the humidity sensor is used to check the humidity in the room. Measurement method is performed by measuring the temperature, and flame and alert through cloud communication. After the detection of fire or smoke, should put-off the fire. For that here a new technology is used. That is a Sonic Fire Extinguisher Extinguish Fire by Sound Waves. Generally, fire is extinguished with the help of water or carbon dioxide. Extinguishing fire through sound bass sounds like crazy. As compared to the other chemical compound of classic extinguishers, this invention offers the cleanest way to put-off flames. Once we predict the fire or smoke by using flame sensor, we will send a data to the ARDUINO UNO micro-controller. With the micro-controller, a SD card is connected and which is storing the sound files. A speaker and amplifier circuit is connected with micro-controller. The controller will read the sound files from SD card and generate the high frequency sound waves to put-off the fire

CHAPTER – 1

INTRODUCTION

A warehouse is a mercantile architecture for entrepot of stuff. Warehouses are used by producers, dealers, traders, wholesalers, distributors, customs, etc. warehouse should be screened at regular intervals to reduce storage cost of food grains due to atmospheric conditions and are documented. With the enlargement of business and the continuous requirements of the food product multiplicity, old style granary/product management prototype will not meet that, due to its heavy capacity and low proficiency. The problem faced by the Central Warehouse Corporation is storage loss of food grains due to environmental changes. We use various sensors to measure the fire/heat of the Warehouse environment and this system avoids granary/product wastage and also intimated to IOT for further actions.

PROBLEM IDENTIFICATION

Hundreds of people who had spent days trapped by fires along a beach in the town of Mallacoota reached safety more than 300 miles away on Saturday morning, after a 20-hour trip on a naval ship.

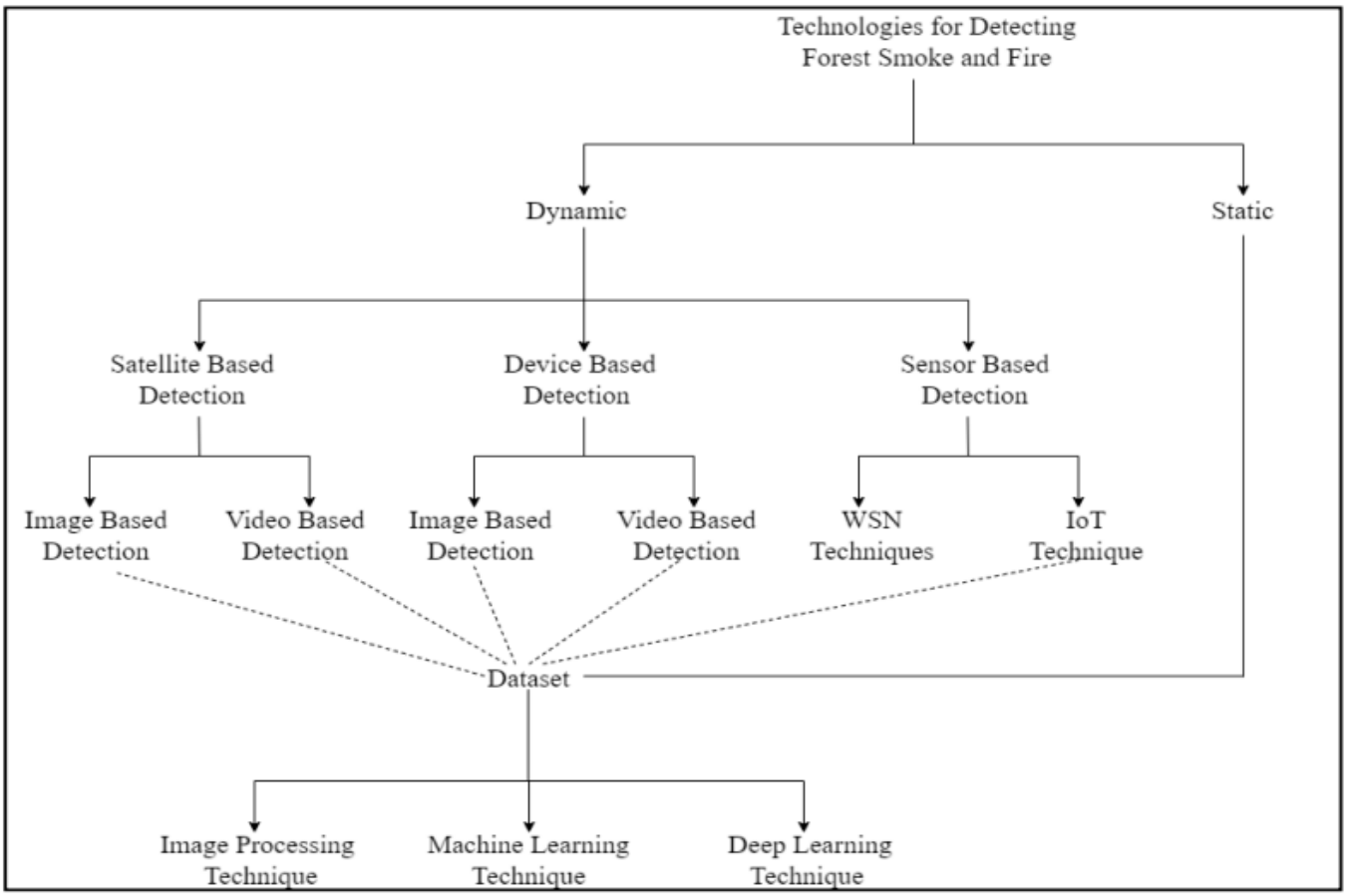
Others had stayed behind, even as Australian officials across three states urged anyone who could leave to do so. By Saturday, numerous towns along Australia's eastern and southeastern coasts were ringed by fire.

This is already one of the worst wildfire seasons Australia has ever endured, and by all measures, Saturday was expected to be even more extreme. High winds and temperatures over 100 degrees Fahrenheit were likely to exacerbate fires already raging out of control. Officials in the state of New South Wales said they expected to lose more houses over the weekend.

OBJECTIVE

- The main goal of the project is to develop advanced concepts of early warehouse fire detection and fire extinguishers.
- An automatic prediction of warehouse fire using machine learning.
- Extinguishes the fire using acoustic waves

TECHNOLOGIES USED FOR DETECTING SMOKE AND FIRE



Warehouse is a planned space for the storage and handling of goods and material. Mostly, ware house are used by manufacturers, importers, exporters, wholesalers etc. It is large commercial building which is located in industrial areas of cities, town and villages. Warehouses are designed for loading and unloading goods directly from airports, seaports

and railways etc. In these days, most of the work is automated with the help of computer and laptop with latest technologies. Efficient monitoring of temperature, humidity and other conditions without being present physically at the location helps us to get a better outcome. Here our main purpose is to observe, control and monitor the warehouse atmosphere, thus making the user to manage the data in real time. Warehouses or storage areas with the small scale units are very close to each other are the leaf nodes of network. They are responsible to collect information about light, temperature and other environmental factors to prevent the food from decaying and getting rotten. Here the central node which is a web application is responsible for passing information to management mode using laptop or mobile phone. Cloud computing is means the practice of using a network of remote servers hosted on the internet of store, manage and process data. Here, cloud computing server stores the data received from various kind of sensors and also send it to farmer mobiles. Internet of things means it is the network of physical devices , vehicles , home appliances and other items embedded with electronics, software , sensors , actuators and connectivity which enables these objects to connect and exchange data.

Digital Warehousing

Digital Warehouse Suite has the capabilities to collect and integrate disparate sources of data within warehouse or distribution facilities: Here, system has capabilities to collect information from various sensors like temperature, humidity, light and smoke sensors and integrate it together.

1. IoT sensor technology to capture and collect real time agricultural/electronic products data. IoT sensors are collecting information's of various parameters from warehouse and uploading it into cloud computing server using IoT.

2. Owner of agricultural products getting information from cloud computing server.

3. This is real time monitoring system of agricultural products.

INTERNET OF THINGS IN WAREHOUSE

Nowadays people are very concerned about the environment because of the rapid changes in the environment which will harm to human health. Hence it is necessary to monitor environment where the people spend more time like at home, office, industry, any working area in real time and long term manner. Using internet of things we can control system as well as we can access system remotely using IoT. It first takes information with help of different sensors and transfer sensors values on cayenne directly, from which can be accessed at anytime and anywhere. Literature survey is done on use of wireless sensors, Cloud and Internet of things, and connection between devices with sensors and network connection will read sensor value which can be further monitored from the internet with the help of cayenne. Monitoring environment is done through website & controlled manually and automatically by detecting sensor values. We can control it manually through website and it can automatically controlled by sensing values. The main Objective design of cloud storage environment is used to store data and to process the data. Internet of things allows physical devices or things which are not computer system, that only act very smartly and makes collaborations decision which are beneficial for different applications. That application allows things to capture value of devices. They transfer “things from being passively computing” and makes an individually decisions in active manner and communicate and collaborate to form single difficult decision. IoT technologies of computing, embedded sensors, communication protocol and internet protocol for communication allow internet of things to provide significant which impose number of challenges and introduces standards which require to specialize and communication.

Forests are the protectors of earth’s ecological balance. Unfortunately, the forest fire is usually only observed when it has already spread over a large area, making its control and stoppage arduous and even impossible at times. The result is devastating loss and irreparable damage to the environment and atmosphere (30% of carbon dioxide (CO2) in the atmosphere

comes from forest fires), in addition to irreparable damage to the ecology (huge amounts of smoke and carbon dioxide (CO2) in the atmosphere). Among other terrible consequences of forest fires are long-term disastrous effects such as impacts on local weather patterns, global warming, and extinction of rare species of the flora and fauna.

Fast and effective detection is a key factor in forest fire fighting. To avoid uncontrollable wide spreading of forest fires it is necessary to detect fires in an early state and to prevent the propagation. It is important to move adequate fire equipment and qualified operational manpower as fast as possible to the source of the fire. Furthermore an adequate logistical infrastructure for sufficient supply with extinguishing devices and maintenance is necessary as well as continuous monitoring of fire spread. Moreover the training of personnel is an important component for successful combating of forest fires. An integrated approach for forest fire detection and suppression is based on a combination of different detection systems depending on wildfire risks, the size of the area and human presence, consisting of all necessary parts such as early detection, remote sensing techniques, logistics, and training by simulation, and fire-fighting vehicles.

Different risk levels, the size of the area and human presence define the applied sensing techniques. Small high risk areas can be observed by local staff. For very large and low risk areas satellite and aero monitoring is possible. Especially in the eastern part of Germany several hundred observation towers equipped with camera-based systems have been setup to observe forests. Recorded image sequences are transmitted to a control center and analyzed by appropriate software. If a fire is clearly identified, fire suppression is initialized by an alarm going directly to the fire brigade. Nowadays, Wireless Sensor Networks (WSNs) are critical components of the increasingly common IoT (Internet of Things) systems. Such systems have a large applicability, and the environmental monitoring field can also benefit from their innovation. The purpose of the IoT concept is to transform the real world and every day electronic devices, appliances, etc., into intelligent

interconnected virtual objects. By keeping the user informed on the state of things and giving the users control of things, a better global humans-devices-humans communication can be achieved.

Forests are the protectors of earth's ecological balance. Unfortunately, the forest fire is usually only observed when it has already spread over a large area, making its control and stoppage arduous and even impossible at times. The result is devastating loss and irreparable damage to the environment and atmosphere (30% of carbon dioxide (CO2) in the atmosphere comes from forest fires), in addition to irreparable damage to the ecology (huge amounts of smoke and carbon dioxide (CO2) in the atmosphere). Among other terrible consequences of forest fires are long-term disastrous effects such as impacts on local weather patterns, global warming, and extinction of rare species of the flora and fauna.

Even in last year a major fire accident occurred in Australia. As fires continue to rip through Australia, some devastating numbers are emerging. At least 24 people killed. More than 15.6 million acres torched. Over 1,400 homes destroyed. And, according to one biodiversity expert's count, an estimated 1 billion animals killed.



Fig 1.1 Forest fire in Australia

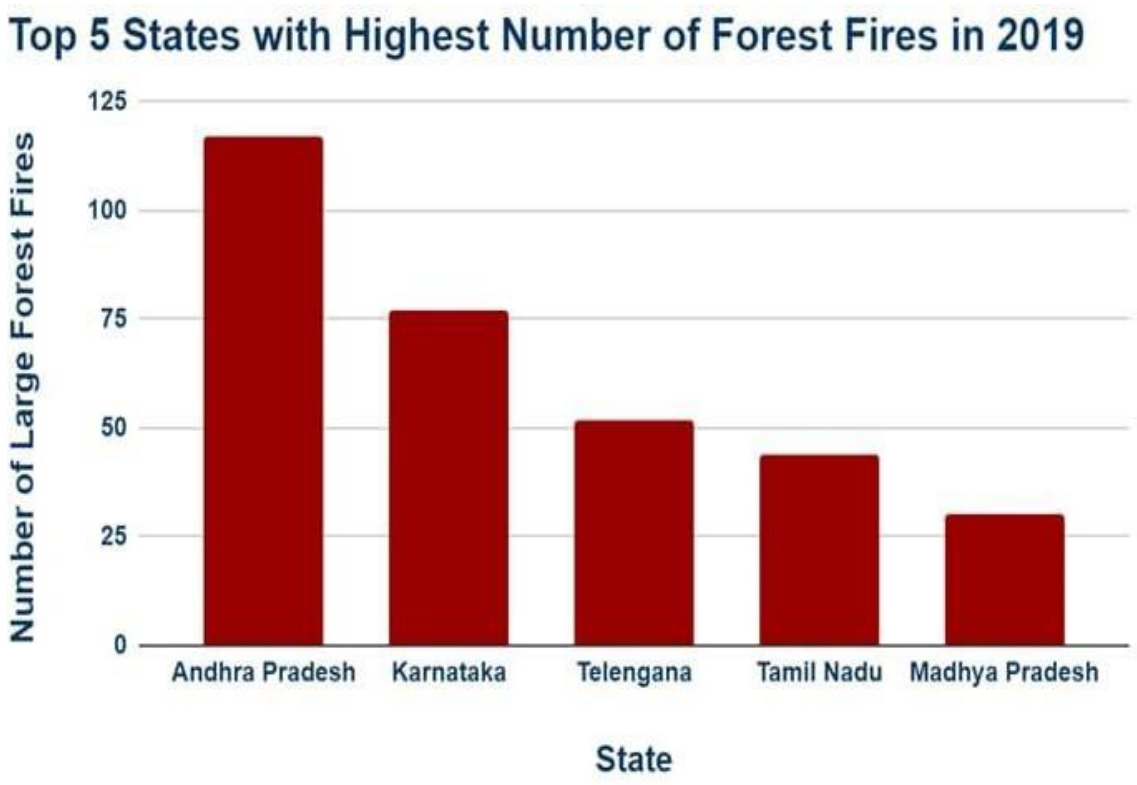


Fig 1.2 Survey of forest fire in 2019

This is the recent survey of major forest fire accident in india. So,the main goal of the project is to develop advanced concepts of early forest fire detection and extinguishers. .In last some years the sensor system and IOT[1] is used to verify an ambiguous situation detected by a video-based system as well as observing an extinguished fire. Therefore sensors have to be widely immune against disturbances like steam, fog, dust pollution and condensing water which usually cause video-based systems to give false alarms. If fire gases are carried to the detector by the airflow they are analyzed with different semiconductor gas sensors. A gas permeable protective cap made of sintered metal protects the sensor elements against soiling with dust and humidity. Thus the sensor array is not

affected by nuisance aerosols like dust, dirt, mist or condensing water . Early forest fire detection sensors have to fulfill a lot of specific requirements compared to conventional applications. High sensitivity is needed to detect even low smoke concentration; dilution and extreme turbulence caused by wind are essential factors due to high occurrence of hydrogen during an open fire a H2-Sensor. So we provide an automatic prediction of forest fire using advanced Artificial intelligence technology[4].

EARLY FIRE DETECTION WITH GAS SENSORS AND SMOKE DETECTORS

The sensor system is used to verify an ambiguous situation detected by a video-based system as well as observing an extinguished fire. Therefore sensors have to be widely immune against disturbances like steam, fog, dust pollution and condensing water which usually cause video-based systems to give false alarms. If fire gases are carried to the detector by the airflow they are analyzed with different semiconductor gas sensors. A gas permeable protective cap made of sintered metal protects the sensor elements against soiling with dust and humidity. Thus the sensor array is not affected by nuisance aerosols like dust, dirt, mist or condensing water [4]. Early forest fire detection sensors (Fig. 6) have to fulfill a lot of specific requirements compared to conventional applications. High sensitivity is needed to detect even low smoke concentration; dilution and extreme turbulence caused by wind are essential factors. Due to high occurrence of hydrogen during an open fire a H2-Sensor

MODEL OF THE PLATFORM FOR EARLY FOREST FIRE DETECTION

Fire detection systems for outdoor environment could be implemented by using specialized cameras, which are able to capture multispectral images. The biggest challenge that arises in these setups is where to place the camera(s) in order to have the best view on the observed territory. Since these systems have their limitations, since they provide stationary point of view, we have decided to investigate a new approach. The platform that is proposed in this paper will use unmanned aerial vehicles, which are going to patrol above the desired territory and will constantly observe for fire-related events. The drones will be equipped with specialized optical and thermal cameras and will be able to capture video or still images.

As stated by National Institute of Disaster Management, Ministry of Home Affairs in their latest report on Forest Fire Disaster Management, forest fire is the major cause of injury and loss to forest. This loss due to fire has a major impact on forest ecosystem their by indirectly affecting the nature's ecosystem. In general fires that occur in the forest can be classified into three types which are:

1. Ground fires,

2. Surface fires, and

3. Crown fires

Ground fires occurs basically on the floor of forest which will produce much heat but without flames. This type of fire is a result of peaty leaves which will be always found on the floor of forest. One more cause can be the organic component of soil which will be formed by the process of decomposition of leaves and other plant materials by soil microorganisms. Ground type of fires is rarest of the three and has been rarely recorded because they normally occur at forests which are situated at very high altitudes such as Himalayan forests. To detect this type of fires, which is a very difficult task, sensors which can record and measure even a temperature difference of as small as 1 °C also should be used. Therefore, one can go for the implementation of thermal sensors and radiation sensors for this purpose.

Surface fires occur on the ground and the spread of this type will always take a regular shape and will usually depend on the speed of wind. To detect this type of fire which will not only produce flames but will also generate smoke one can go for smoke sensor. One such use of smoke to detect fire in home environment is dis-cussed. In this article, authors have given importance to the optimization of power in terms of both the hardware and software. Lastly, the crown type of fires will burn the complete tree right from the root to the top most

leaf through the stem. This type of fires will give much more flame then smoke. To detect these fires one needs a very robust sensor that can withstand a very high temperature.

CHAPTER – 2

LITERATURE SURVEY

1) **Ali Fahem Neamah et al "Flexible Data Warehouse: Towards Building an Integrated Electronic Health Record Architecture" – IEEE 2020.** In the data warehouse based implementation, the records about the particular organization or different organizations are stored to be fetched for future usage. Electronic health record (EHR) is an important system of information and communication technologies to the healthcare sector. EHR implementation is expected to produce benefits for patients, professionals, organizations, and the population as a whole.
2) Emir Cogo Emir Žunić Admir Beširević et al "**Position based visualization of real world warehouse data in a smart warehouse management system – IEEE, 2020.** Data visualization method in 3D space that includes actual positions, volumes and space relations of the chunks of data that are being visualized. Data that is being visualized is real-time information provided by the smart warehouse management system about packages distributed on pallet places within a warehouse.
3) Mohamed Dhouioui et al "**Intelligent Warehouse Management System" – IEEE, 2020.** An robot that moves products in the warehouse according to storage and shipping requests. Our solution is designed to allow the various actors to have real-time information on the different workflows within the warehouse and all the movements of stock. Hence the need for a system that controls all zones and locations and ensures communication between the various actors and software components while optimizing data exchange and load consumption for lOT equipment.
4) **K.Mohanraj, S.Vijayalakshmi, N.Balaji, R.Chithrakkannan, R.Karthikeyan et al "Smart Warehouse Monitoring Using IoT" IJEAT – 2019.** sensors includes

vibration, humidity, temperature, fire sensors. It is done with the help of current technology (IoT). Raspberry pi controller adopts IoT technology to convey the messages. Based on the sensor's data the appropriate data is captured and manipulated based on the limit given in the software and send timely information to the concern department officials of Central warehouse corporation through SMS for moderation and corrective actions arising due to atmospheric conditions inside the warehouse.

5) **Tejas S Patil, Sagar R Khedkar, Mahesh S Jadhav et al "Internet of Things (IoT) based Warehouse Monitoring and Control Interface Implementation" – IRJET, 2019.** system solution monitors temperature, humidity and various gases like LPG, CO, and Methane, whereas also provide the location coordinates of the warehouse in real time to keep food, vegetables, medicines, and vaccines safe in addition to reduce wastage. This IoT compromises of most advanced sensors and Particle cloud software. The sensor data is sent from node to the base station then will process and displayed on LCD. The data is pushed and stored on particle cloud and can be accessed remotely using a mobile application Blynk.
6) **T. Saikumar, P. Sriramya et al "Iot Enabled Forest Fire Detection and Altering the Authorities" – IJRTE, 2019**

This system for tracking and alarming for the protection of trees against forest fires. Nowadays IOT (Internet of Things) devices and sensors allow the monitoring of different environmental variables, such as temperature, humidity, moisture etc. Arduino platform based IOT enabled fire detector and monitoring system is the solution to this problem. In this project we have built fire detector using Arduino UNO which is interfaced with a temperature sensor, a smoke sensor and a buzzer. In order to implement this project, we will be using GSM which is used to provide the final SMS to the user through the given number in the simulation program, Temperature sensor which is used to denote the temperature High and Low that will be displayed in the LCD Display, Flame sensor which is used to denote

the flame ranges and if it is high the forest fire will be detected in the LCD display and if it is low forest fire won't be detected. Whenever a fire occurs, the system automatically senses and alerts the user by sending an alert to an app installed on user's android mobile or webpage accessible through the internet.

7) **Aditi Kansal et al "Detection of Forest Fires using Machine Learning Technique: A Perspective" – IEEE conference – 2015.**

Detection of these disasters should be fast and accurate as they may cause damage and destruction at a large scale. In this paper, comparison of various machine learning techniques such as SVM, regression, decision trees, neural networks etc. has been done for prediction of forest fifes. The proposed approach in this paper presents how regression works best for detection of forest fires with high accuracy by dividing the dataset. Fast detection of forest fires is done in this paper by taking less time as compared to other machine learning techniques.

8) **Alina-Elena Marcu et al "IoT System for Forest Monitoring", IEEE – 2019**

This system designed a forest environment monitoring solution based on the Raspberry Pi Model 3, analogical and digital sensors and signals analysis algorithms. Parameters such as temperature, gas concentrations, soil humidity etc. are monitored with sensors while background sounds are analyzed with a classification algorithm on the basis of which the generated event can be classified into one of the following categories: Chainsaw, Vehicle, or Forest background noise. The user's accessibility to the collected data is ensured via Internet and a mobile applications that allows the user to receive notifications, whenever fire, pollution sources, or illegal deforestation are detected. The SeaForest environment monitoring solution is an IoT project, addressed to public and private forest owners as well as to national environmental and disaster response authorities.

9) **Semisupervised Classification Based Clustering Approach in WSN for Forest Fire Detection – Research gate , 2019**

A semisupervised rule-based classification model is proposed in this paper to detect whether its zone is high active, medium active (MA) or low active (LA) cluster in the forest. We train our proposed integrated model in such a way when only one parameter of sensed data is transmitted by the sensor nodes due to energy constraint to the initiator of that zone, initiator can be able to predict the state of (HA,MA,LA) zone with 96% accuracy. All the sensor nodes in HA cluster transmit their packet through cluster head to the base station continuously applying greedy forwarding technique. Authors consider energy saving strategy during cluster head selection and data transmission in HA zone. On the other hand, sensors in MA zone transmit packet periodically and LA zone avoids to transmit the sensed data. This way proposed technique transmits the sensed data from HA zone efficiently and quickly to forest office for forest fire prevention and saves the energy of all sensor nodes in the forest.

CHAPTER – 3

SYSTEM DESIGN

EXISTING SYSTEM

- Now a day's storehouses need a low operating cost technology hence, required minimum managers for efficient operation of the storehouse management administration. Let's discuss those new modern technologies:
- A.WMS: In the last decades, more advanced technologies used in the storehouse management systems. Labour-intensiveness is reduced due to efficient and time consuming process.
- B. Data entry and Paperwork: Data entry and paper work has reduced the time working with the spread sheets and ledger maintenance of the management system.
- C. Selection efficiency: With the help of computer guided system the operatives can work faster with WMS, because the new modern technology helps us to arrange systematically for efficient and real time management of the WMS.
- D. Task Interleaving: It becomes more powerful tools so that system guidance will be extended to all the activities. Especially it is used for the operator of forklift.

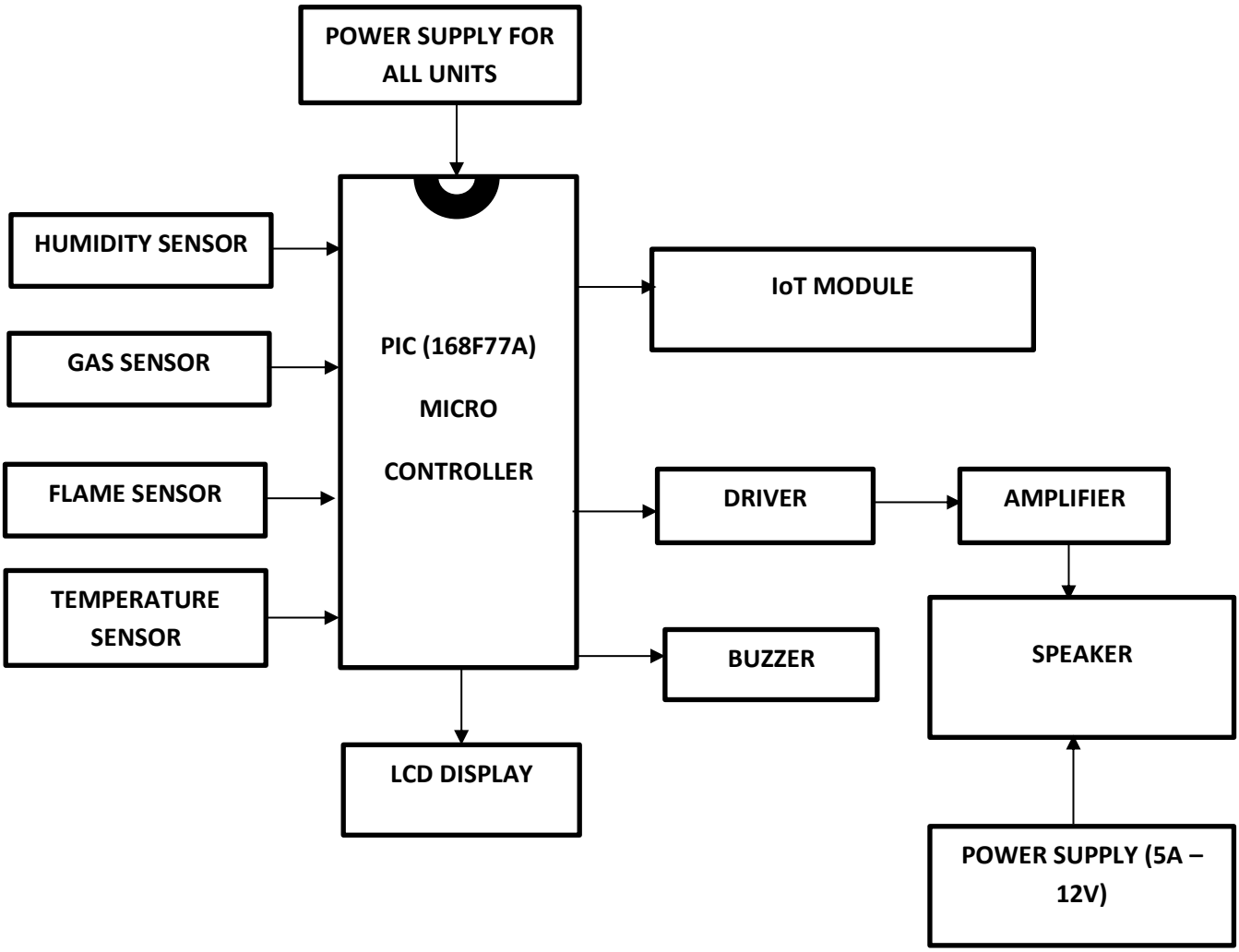
Limitations:

- The data lacks precise ignition point information.
- Countries with poor infrastructure are often unable to conduct their own field surveys which make it difficult to obtain basic historical data on previous fire sources, thus restricting their accurate prediction.

PROPOSED SYSTEM

- Nature has a huge role to maintain the stability in the environment. But, natural disasters damage the environment, affect the life cycle and decline the lifetime of living beings. Nature is destroyed by different disasters namely earthquake, fire, flood, landslide, air pollution and so on.
- Warehouse/Forest fire is one of the foremost dangerous, which cause several serious issues like biodiversity loss, global warming, fuel wood loss and air pollution in the environment. Therefore, prediction of fire occurring in warehouse/forest plays a crucial role to save lots of the environment.
- Gas, flame and temperature are the focal point for fire, some of the researchers pay their attention on detecting the smoke of the forest using different technologies. In this proposed system for effectively detecting the smoke and fire in the warehouse/forest. Gas-based fire detection could provide an additional level of safety to building occupants.
- Sensor based technology is also used for detecting the occurrence of smoke and fire in warehouse. IoT devices have different sensors that are deployed in the device itself to generate the alarming only if there is any deflection in the environment.

BLOCK DIAGRAM



CHAPTER – 4

HARDWARE DESCRIPTION

POWER SUPPLY

Power supply is a reference to a source of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

Power supplies for electronic devices can be broadly divided into linear and switching power supplies. The linear supply is a relatively simple design that becomes increasingly bulky and heavy for high current devices; voltage regulation in a linear supply can result in low efficiency. A switched-mode supply of the same rating as a linear supply will be smaller, is usually more efficient, but will be more complex.

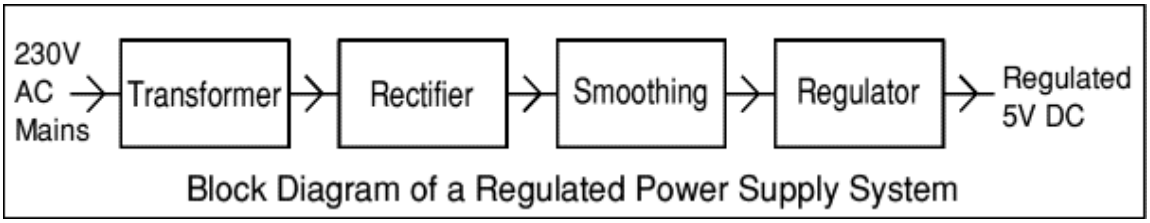
Linear Power supply:

An AC powered linear power supply usually uses a transformer to convert the voltage from the wall outlet (mains) to a different, usually a lower voltage. If it is used to produce DC, a rectifier is used. A capacitor is used to smooth the pulsating current from the rectifier. Some small periodic deviations from smooth direct current will remain, which is known as ripple. These pulsations occur at a frequency related to the AC power frequency (for example, a multiple of 50 or 60 Hz).

The voltage produced by an unregulated power supply will vary depending on the load and on variations in the AC supply voltage. For critical electronics applications a linear regulator will be used to stabilize and adjust the voltage. This regulator will also greatly

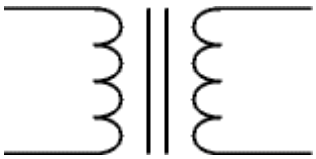
reduce the ripple and noise in the output direct current. Linear regulators often provide current limiting, protecting the power supply and attached circuit from over current.

Adjustable linear power supplies are common laboratory and service shop test equipment, allowing the output voltage to be set over a wide range. For example, a bench power supply used by circuit designers may be adjustable up to 30 volts and up to 5 amperes output. Some can be driven by an external signal, for example, for applications requiring a pulsed output.



Block Diagram of a Regulated Power Supply System

Transformer:



Transformers convert AC electricity from one voltage to another with little loss of power. Transformers work only with AC and this is one of the reasons why mains electricity is AC.

Step-up transformers increase voltage, step-down transformers reduce voltage. Most power supplies use a step-down transformer to reduce the dangerously high mains voltage (230V in UK) to a safer low voltage.

The input coil is called the primary and the output coil is called the secondary. There is no electrical connection between the two coils; instead they are linked by an alternating

magnetic field created in the soft-iron core of the transformer. The two lines in the middle of the circuit symbol represent the core.

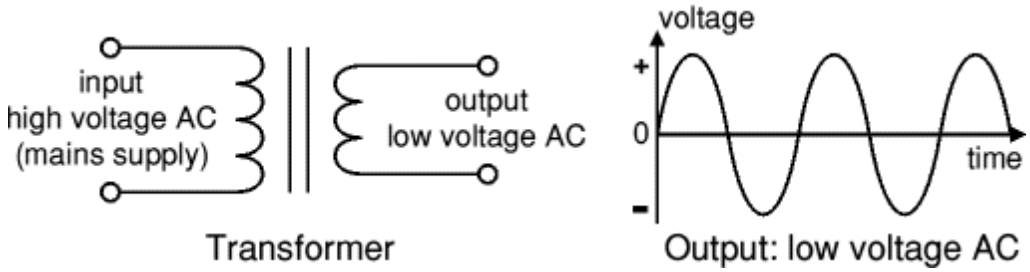
Transformers waste very little power so the power out is (almost) equal to the power in. Note that as voltage is stepped down current is stepped up.

The ratio of the number of turns on each coil, called the turn's ratio, determines the ratio of the voltages. A step-down transformer has a large number of turns on its primary (input) coil which is connected to the high voltage mains supply, and a small number of turns on its secondary (output) coil to give a low output voltage.

Turns ratio=Vp/Vs=Nn/Ns and Power out=Power in

Vs*Is=Vp * Ip

Vp = primary (input) voltage Vs = secondary (output) voltage
Np = number of turns on primary coil Ns = number of turns on secondary coil
Ip = primary (input) current Is = secondary (output) current

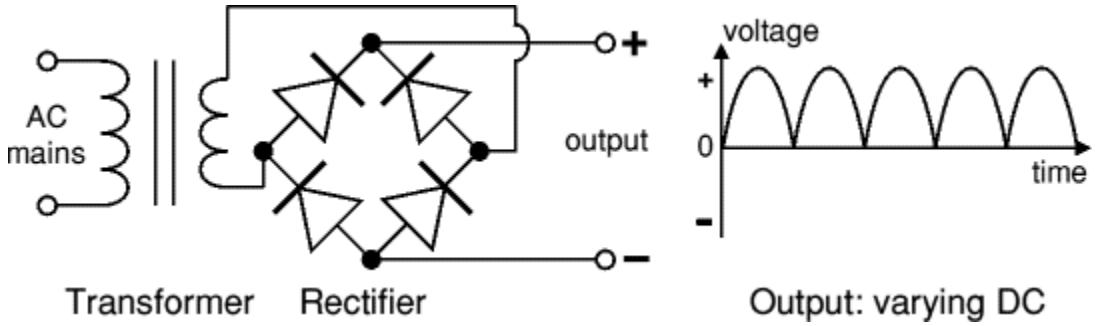


Transformer Output: low voltage AC

The low voltage AC output is suitable for lamps, heaters and special AC motors. It is not suitable for electronic circuits unless they include a rectifier and a smoothing capacitor.

Rectifier:

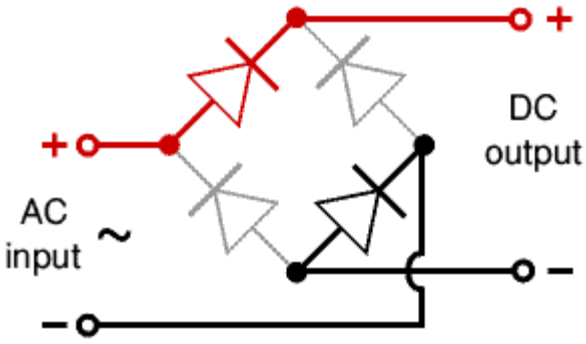
There are several ways of connecting diodes to make a rectifier to convert AC to DC. The bridge rectifier is the most important and it produces full-wave varying DC. A full-wave rectifier can also be made from just two diodes if a centre-tap transformer is used, but this method is rarely used now that diodes are cheaper. A single diode can be used as a rectifier but it only uses the positive (+) parts of the AC wave to produce half-wave varying DC.



The varying DC output is suitable for lamps, heaters and standard motors. It is not suitable for electronic circuits unless they include a smoothing capacitor.

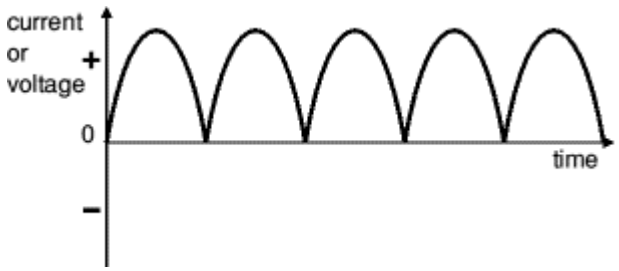
Bridge rectifier:

A bridge rectifier can be made using four individual diodes, but it is also available in special packages containing the four diodes required. It is called a full-wave rectifier because it uses the entire AC wave (both positive and negative sections). 1.4V is used up in the bridge rectifier because each diode uses 0.7V when conducting and there are always two diodes conducting, as shown in the diagram below. Bridge rectifiers are rated by the maximum current they can pass and the maximum reverse voltage they can withstand (this must be at least three times the supply RMS voltage so the rectifier can withstand the peak voltages). Please see the Diodes page for more details, including pictures of ridge rectifiers.



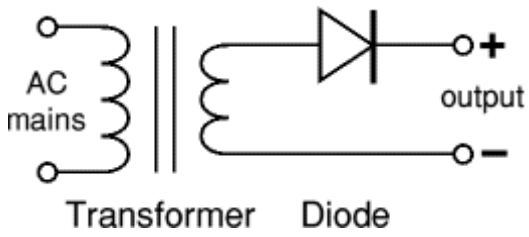
Alternate pairs of diodes conduct, changing over the connections so the alternating directions of AC are converted to the one direction of DC.

Output: full-wave varying DC: (using the entire AC wave):

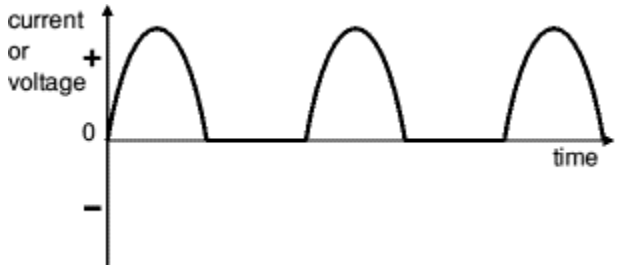


Single diode rectifier:

A single diode can be used as a rectifier but this produces **half-wave** varying DC which has gaps when the AC is negative. It is hard to smooth this sufficiently well to supply electronic circuits unless they require a very small current so the smoothing capacitor does not significantly discharge during the gaps. Please see the Diodes page for some examples of rectifier diodes.

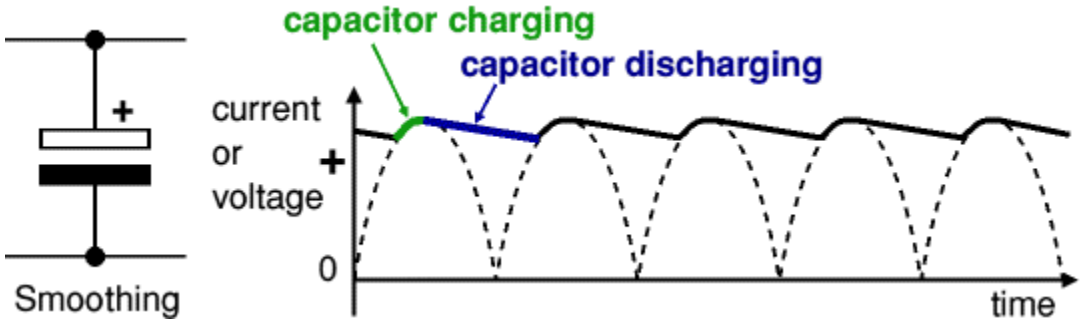


Output: half-wave varying DC (using only half the AC wave):



Smoothing:

Smoothing is performed by a large value electrolytic capacitor connected across the DC supply to act as a reservoir, supplying current to the output when the varying DC voltage from the rectifier is falling. The diagram shows the unsmoothed varying DC (dotted line) and the smoothed DC (solid line). The capacitor charges quickly near the peak of the varying DC, and then discharges as it supplies current to the output.



Note that smoothing significantly increases the average DC voltage to almost the peak value (1.4 × RMS value). For example 6V RMS AC is rectified to full wave DC of about 4.6V RMS (1.4V is lost in the bridge rectifier), with smoothing this increases to almost the peak value giving 1.4 × 4.6 = 6.4V smooth DC.

Smoothing is not perfect due to the capacitor voltage falling a little as it discharges, giving a small ripple voltage. For many circuits a ripple which is 10% of the supply voltage is satisfactory and the equation below gives the required value for the smoothing capacitor. A larger capacitor will give fewer ripples. The capacitor value must be doubled when smoothing half-wave DC.

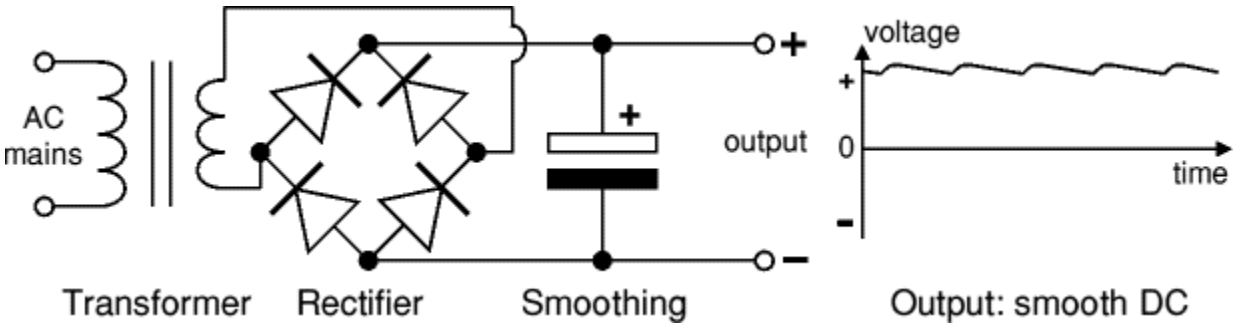
Smoothing Capacitor for 10% ripple, C=5*10/vs.*f

C = smoothing capacitance in farads (F)

Io = output current from the supply in amps (A)

Vs = supply voltage in volts (V), this is the peak value of the unsmoothed DC

f = frequency of the AC supply in hertz (Hz), 50Hz in the UK.



The smooth DC output has a small ripple. It is suitable for most electronic circuits.

Regulator:

Voltage regulator ICs are available with fixed (typically 5, 12 and 15V) or variable output voltages. They are also rated by the maximum current they can pass. Negative voltage regulators are available, mainly for use in dual supplies. Most regulators include some automatic protection from excessive current ('overload protection') and overheating ('thermal protection').

The LM78XX series of three terminal regulators is available with several fixed output voltages making them useful in a wide range of applications. One of these is local on card regulation, eliminating the distribution problems associated with single point regulation. The voltages available allow these regulators to be used in logic systems, instrumentation, HiFi, and other solid state electronic equipment. Although designed primarily as fixed voltage regulators these devices can be used with external components to obtain adjustable voltages and current.

Many of the fixed voltage regulator ICs has 3 leads and look like power transistors, such as the 7805 +5V 1A regulator shown on the right. They include a hole for attaching a heat sink if necessary.

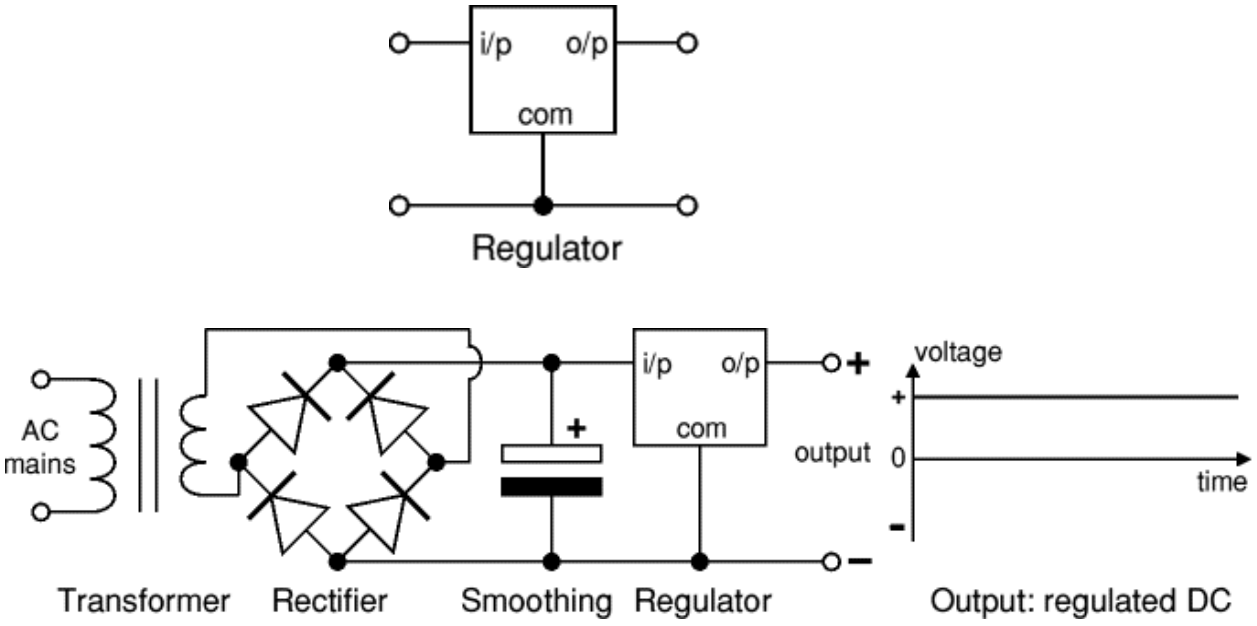
1. Positive regulator
    1. input pin
    2. ground pin
    3. output pin

It regulates the positive voltage

2. Negative regulator
    1. ground pin
    2. input pin

3. output pin

It regulate the negative voltage

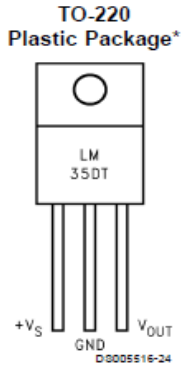


The regulated DC output is very smooth with no ripple. It is suitable for all electronic circuits.

TEMPERATURE SENSOR (LM35):

The LM35 is an integrated circuit sensor that can be used to measure temperature with an electrical output proportional to the temperature (in ºC). The LM35 generates a higher output voltage than thermocouples and may not require that the output voltage be amplified.

PIN DIAGRAM:



DESCRIPTION

- It has an output voltage that is proportional to the Celsius temperature.
- The scale factor is .01V/ºC
- The LM35 does not require any external calibration or trimming and maintains an accuracy of +/-0.4 ºC at room temperature and +/- 0.8 ºC over a range of 0 ºC to +100 ºC.
- Another important characteristic of the LM35DZ is that it draws only 60 micro amps from its supply and possesses a low self-heating capability. The sensor self-heating causes less than 0.1 ºC temperature rise in still air.

The LM35 series are precision integrated-circuit temperature sensors, whose output voltage is linearly proportional to the Celsius (Centigrade) temperature. The LM35 thus has an advantage over linear temperature sensors calibrated in ° Kelvin, as the user is not required to subtract a large constant voltage from its output to obtain convenient Centigrade scaling.

The LM35 does not require any external calibration or trimming to provide typical accuracies of ±¼°C at room temperature and ±¾°C over a full -55 to +150°C temperature range. Low cost is assured by trimming and calibration at the wafer level. The LM35's low output impedance, linear output, and precise inherent calibration make interfacing to readout or control circuitry especially easy. It can be used with single

power supplies, or with plus and minus supplies. As it draws only 60 μA from its supply, it has very low self-heating, less than 0.1°C in still air.

The LM35 is rated to operate over a -55° to +150°C temperature range, while the LM35C is rated for a -40° to +110°C range (-10° with improved accuracy). The LM35 series is available packaged in hermetic TO-46 transistor packages, while the LM35C, LM35CA, and LM35D are also available in the plastic TO-92 transistor package. The LM35D is also available in an 8-lead surface mount small outline package and a plastic TO-220 package

Features:

- Calibrated directly in ° Celsius (Centigrade)
- Linear + 10.0 mV/°C scale factor
- 0.5°C accuracy guarantee able (at +25°C)
- Rated for full −55° to +150°C range
- Suitable for remote applications
- Low cost due to wafer-level trimming
- Operates from 4 to 30 volts
- Less than 60 μA current drain
- Low self-heating, 0.08°C in still air
- Nonlinearity only ±1⁄4°C typical
- Low impedance output, 0.1 W for 1 mA load

FLAME SESNOR

A **flame detector** is a sensor designed to detect and respond to the presence of a flame or fire, allowing **flame detection**. Responses to a detected flame depend on the installation, but can include sounding an alarm, deactivating a fuel line (such as a propane or a natural gas line), and activating a fire suppression system. When used in applications such as industrial furnaces, their role is to provide confirmation that the furnace is working properly; it can be used to turn off the ignition system though in many cases they take no direct action beyond notifying the operator or control system. A flame detector can often respond faster and more accurately than a smoke or heat detector due to the mechanisms it uses to detect the flame.

This sensor detects flame otherwise wavelength within the range of 760 nm – 1100 nm from the light source. This sensor can be easily damaged to high temperature. So this sensor can be placed at a certain distance from the flame. The flame detection can be done from a 100cm distance and the detection angle will be 600. The output of this sensor is an analog signal or digital signal. These sensors are used in fire fighting robots like as a flame alarm.

What is a Flame Sensor?

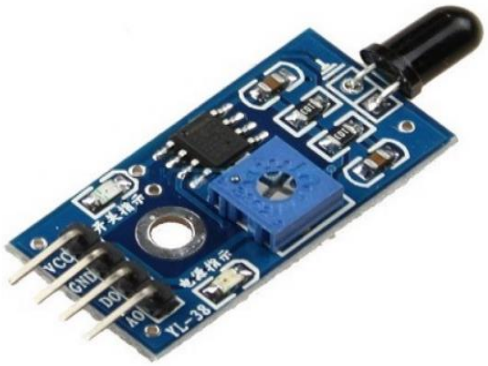
A flame-sensor is one kind of detector which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. The flame detection response can depend on its fitting. It includes an alarm system, a natural gas line, propane & a fire suppression system. This sensor is used in industrial boilers. The main function of this is to give authentication whether the boiler is properly working or not. The response of these sensors is faster as well as more accurate compare with a heat/smoke detector because of its mechanism while detecting the flame.

Working Principle

This sensor/detector can be built with an electronic circuit using a receiver like electromagnetic radiation. This sensor uses the infrared flame flash method, which allows the sensor to work through a coating of oil, dust, water vapor, otherwise ice.

Flame Sensor Module

The pin configuration of this sensor is shown below. It includes four pins which include the following. When this module works with a microcontroller unit then the pins are



Flame Sensor

- Pin1 (VCC pin): Voltage supply rages from 3.3V to 5.3V
- Pin2 (GND): This is a ground pin
- Pin3 (AOUT): This is an analog output pin (MCU.IO)
- Pin4 (DOUT): This is a digital output pin (MCU.IO)

Features & Specifications

The features of this sensor include the following.

- Photosensitivity is high
- Response time is fast
- Simple to use

- Sensitivity is adjustable
- Detection angle is 600,
- It is responsive to the flame range.
- Accuracy can be adjustable
- Operating voltage of this sensor is 3.3V to 5V
- Analog voltage o/ps and digital switch o/ps
- The PCB size is 3cm X 1.6cm
- Power indicator & digital switch o/p indicator
- If the flame intensity is lighter within 0.8m then the flame test can be activated, if the flame intensity is high, then the detection of distance will be improved.

Applications

These sensors are used in several dangerous situations which include the following.

- Hydrogen stations
- Industrial heating
- Fire detection
- Fire alarm
- Fire fighting robot
- Drying systems
- Industrial gas turbines
- Domestic heating systems
- Gas-powered cooking devices

GAS SENSOR

- Electrochemical gas sensors are gas detectors that measure the concentration of a target gas by oxidizing or reducing the target gas at an electrode and measuring the resulting current
- Gas is a colorless, odor and poisonous gas formed when carbon in fuels is not burned completely.
- This semiconductor **gas sensor** detects the presence of **harmful** (CNG) **gas** at concentrations from 300 ppm to 10,000 ppm, a range suitable for detecting sewer waste gas leaks. The **sensor's** simple analog voltage interface requires only one analog input pin from your microcontroller.
- This is a simple-to-use gas sensor suitable for sensing gas concentrations in the air.



Pin Name	Description
VCC	This pin powers the module, typically the operating voltage is +5V

GND	Used to connect the module to system ground
Digital Out (DO)	You can also use this sensor to get digital output from this pin, by setting a threshold value using the potentiometer
Analog Out (AO)	This pin outputs 0-5V analog voltage based on the intensity of the gas

Construction

The sensors contain two or three electrodes, occasionally four, in contact with an electrolyte. The electrodes are typically fabricated by fixing a high surface area precious metal on to the porous hydrophobic membrane. The working electrode contacts both the electrolyte and the ambient air to be monitored usually via a porous membrane. The electrolyte most commonly used is a mineral acid, but organic electrolytes are also used for some sensors. The electrodes and housing are usually in a plastic housing which contains a gas entry hole for the gas and electrical contacts.

Theory of operation

The gas diffuses into the sensor, through the back of the porous membrane to the working electrode where it is oxidized or reduced. This electrochemical reaction results in an electric current that passes through the external circuit. In addition to measuring, amplifying and performing other signal processing functions, the external circuit maintains the voltage across the sensor between the working and counter electrodes for a two electrode sensor or

between the working and reference electrodes for a three electrode cell. At the counter electrode an equal and opposite reaction occurs, such that if the working electrode is an oxidation, then the counter electrode is a reduction.

Diffusion controlled response

The magnitude of the current is controlled by how much of the target gas is oxidized at the working electrode. Sensors are usually designed so that the gas supply is limited by diffusion and thus the output from the sensor is linearly proportional to the gas concentration. This linear output is one of the advantages of electrochemical sensors over other sensor technologies, (e.g. infrared), whose output must be linearized before they can be used. A linear output allows for more precise measurement of low concentrations and much simpler calibration (only baseline and one point are needed).

Diffusion control offers another advantage. Changing the diffusion barrier allows the sensor manufacturer to tailor the sensor to a particular target gas concentration range. In addition, since the diffusion barrier is primarily mechanical, the calibration of electrochemical sensors tends to be more stable over time and so electrochemical sensor based instruments require much less maintenance than some other detection technologies. In principle, the sensitivity can be calculated based on the diffusion properties of the gas path into the sensor, though experimental errors in the measurement of the diffusion properties make the calculation less accurate than calibrating with test gas.

Cross sensitivity

For some gases such as ethylene oxide, cross sensitivity can be a problem because ethylene oxide requires a very active working electrode catalyst and high operating potential for its oxidation. Therefore gases which are more easily oxidized such as alcohols and carbon monoxide will also give a response. Cross sensitivity problems can be eliminated though

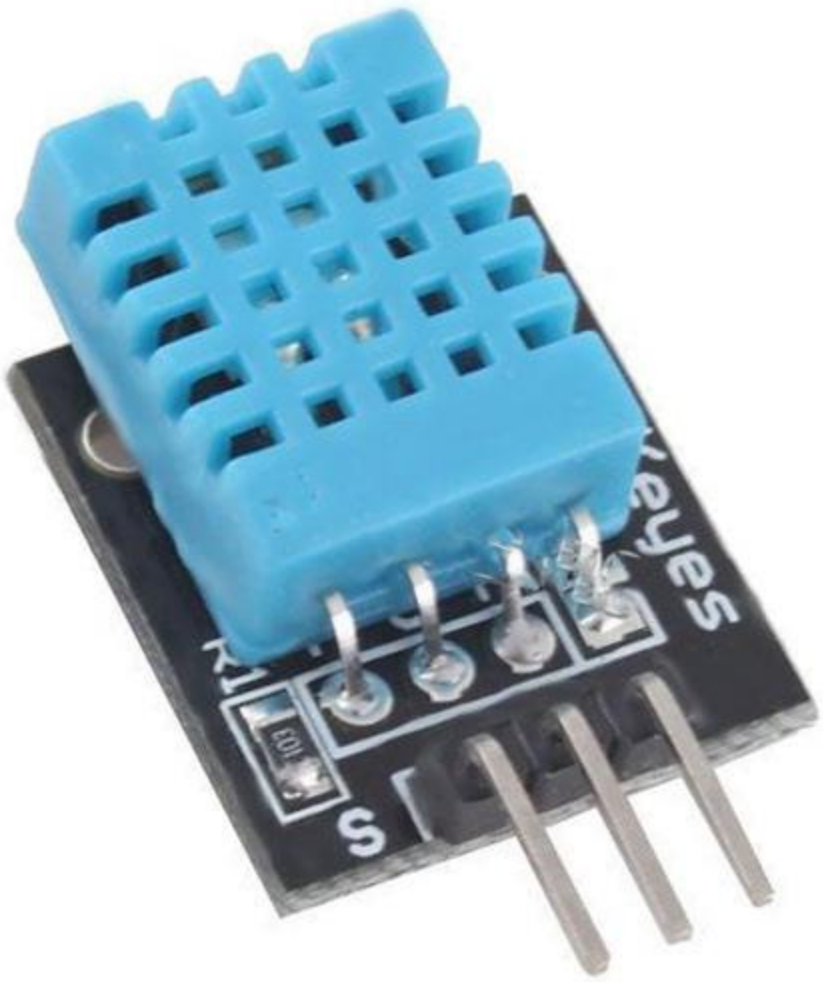
through the use of a chemical filter, for example filters that allows the target gas to pass through unimpeded, but which reacts with and removes common interferences.

While electrochemical sensors offer many advantages, they are not suitable for every gas. Since the detection mechanism involves the oxidation or reduction of the gas, electrochemical sensors are usually only suitable for gases which are electrochemically active, though it is possible to detect electrochemically inert gases indirectly if the gas interacts with another species in the sensor that then produces a response.[2] Sensors for carbon dioxide are an example of this approach and they have been commercially available for several years.

HUMIDITY SENSOR

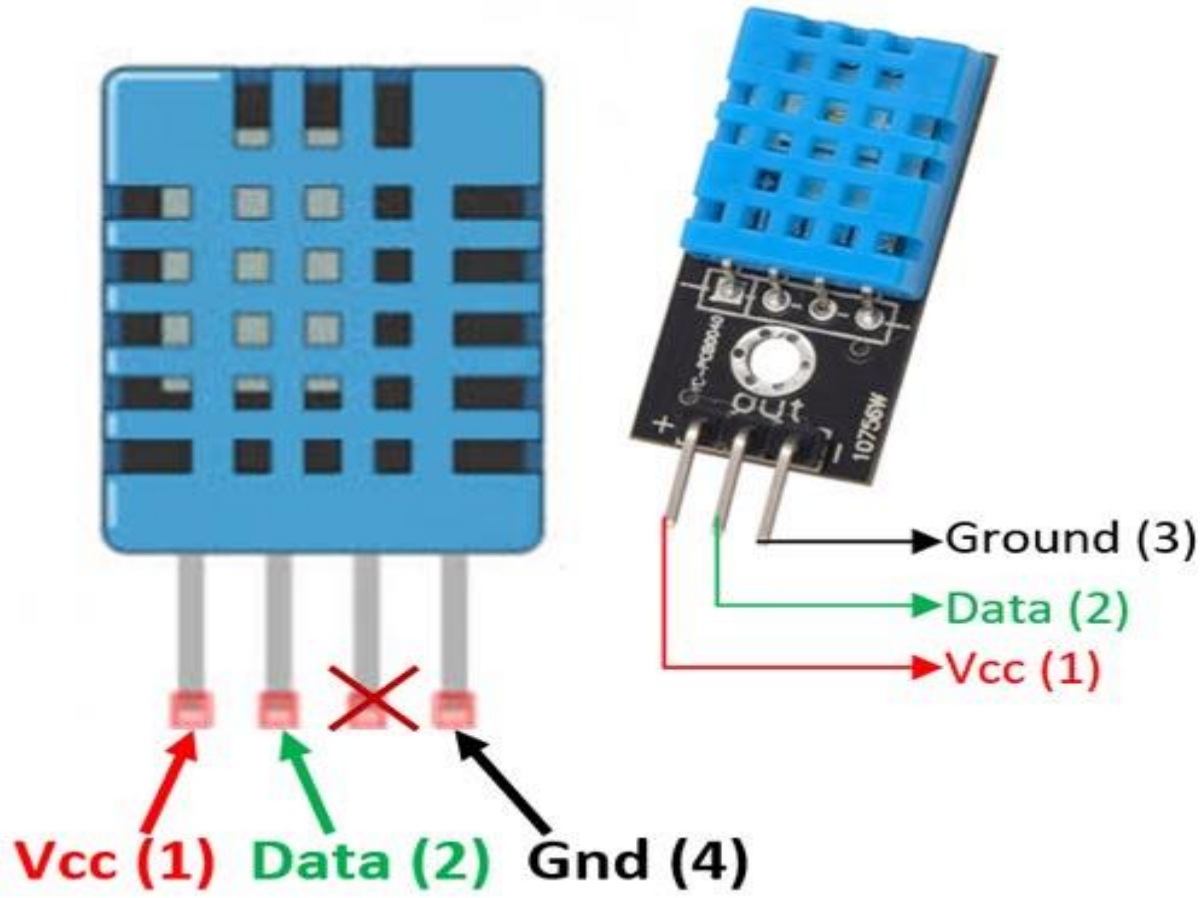
Pin Identification and Configuration:

No:	Pin Name	Description
For DHT11 Sensor		
1	Vcc	Power supply 3.5V to 5.5V
2	Data	Outputs both Temperature and Humidity through serial Data
3	NC	No Connection and hence not used
4	Ground	Connected to the ground of the circuit
For DHT11 Sensor module		
1	Vcc	Power supply 3.5V to 5.5V
2	Data	Outputs both Temperature and Humidity through serial Data
3	Ground	Connected to the ground of the circuit



DHT11 Specifications:

- Operating Voltage: 3.5V to 5.5V
- Operating current: 0.3mA (measuring) 60uA (standby)
- Output: Serial data
- Temperature Range: 0°C to 50°C
- Humidity Range: 20% to 90%
- Resolution: Temperature and Humidity both are 16-bit
- Accuracy: ±1°C and ±1%



DHT11–Temperature and Humidity Sensor DHT11 Sensor Pinout

Difference between DHT11 Sensor and module:

The **DHT11 sensor** can either be purchased as a sensor or as a module. Either way, the performance of the sensor is same. The sensor will come as a 4-pin package out of which only three pins will be used whereas the module will come with three pins as shown above.

The only difference between the sensor and module is that the module will have a filtering capacitor and pull-up resistor inbuilt, and for the sensor, you have to use them externally if required.

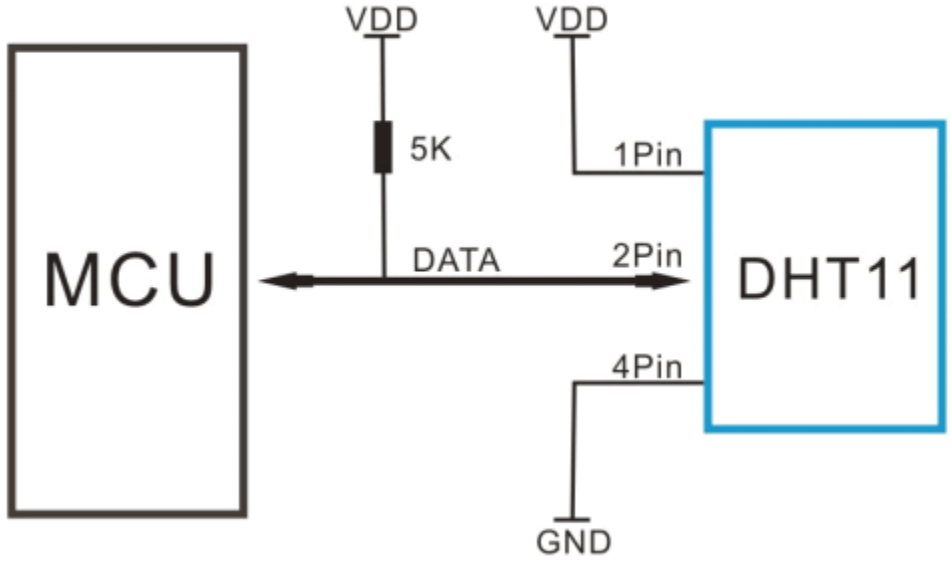
Where to use DHT11:

The **DHT11** is a commonly used **Temperature and humidity sensor.** The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory calibrated and hence easy to interface with other microcontrollers.

The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of ±1°C and ±1%. So if you are looking to measure in this range then this sensor might be the right choice for you.

How to use DHT11 Sensor:

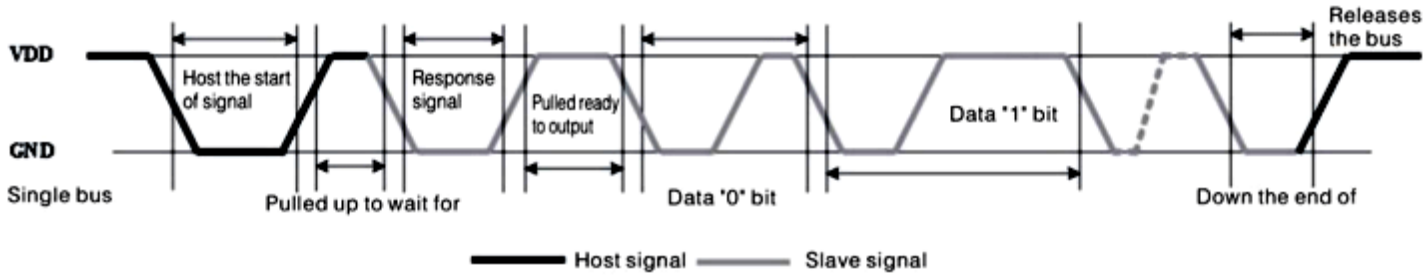
The DHT11 Sensor is factory calibrated and outputs serial data and hence it is highly easy to set it up. The connection diagram for this sensor is shown below.



As you can see the data pin is connected to an I/O pin of the MCU and a 5K pull-up resistor is used. This data pin outputs the value of both temperature and humidity as serial data. If you are trying to interface DHT11 with Arduino then there are ready-made libraries for it which will give you a quick start.

If you are trying to interface it with some other MCU then the datasheet given below will come in handy. The output given out by the data pin will be in the order of 8bit humidity integer data + 8bit the Humidity decimal data +8 bit temperature integer data + 8bit fractional temperature data

+8 bit parity bit. To request the DHT11 module to send these data the I/O pin has to be momentarily made low and then held high as shown in the timing diagram below

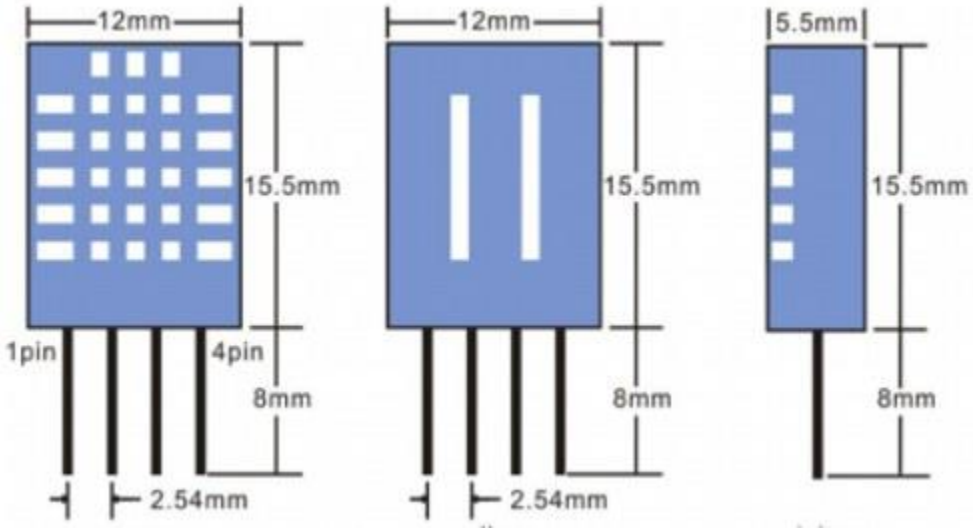


The duration of each host signal is explained in the DHT11 datasheet, with neat steps and illustrative timing diagrams

Applications:

- Measure temperature and humidity
- Local Weather station
- Automatic climate control
- Environment monitoring

2D–model of the sensor:



SD CARD MODULE

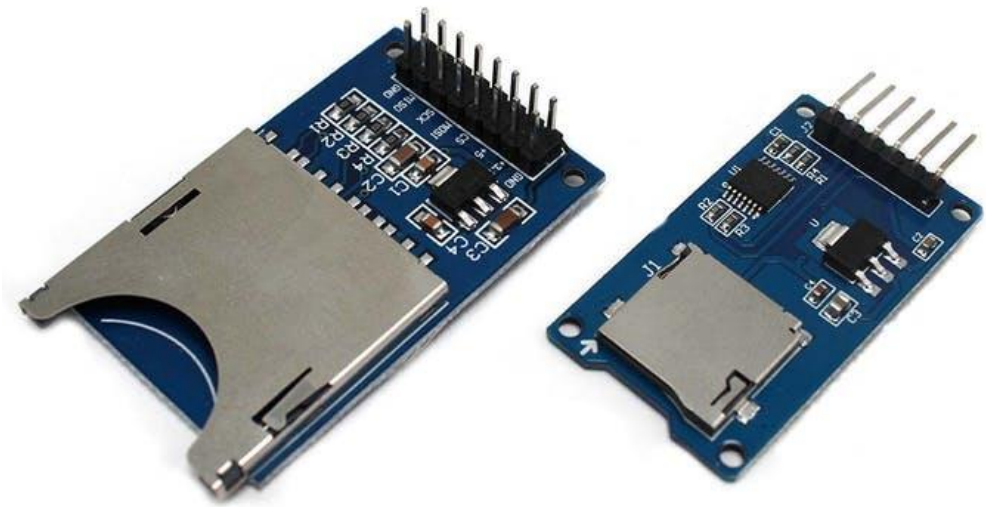
The SD and micro SD card modules allow you to communicate with the memory card and write or read the information on them. The module interfaces in the SPI protocol. In this project, 30 Hz – 60Hz frequency is stored in the SD card to put off the fire.

To use these modules with Arduino you need the SD library. This library is installed on the Arduino application by default.

Description

The Micro SD Card Module has SPI interface which is compatible with any sd card and it use 5V or 3.3V power supply which is compatible with most micro controllers boards.

SD module has various applications such as data logger, audio, video, graphics. This module will greatly expand the capability an Arduino can do with their poor limited memory.



Micro SD Card Module Specifications

- Working Voltage: 5V/3.3V
- Size:20x28mm
- Interface: SPI

- Compatible: MicroSD

Since Arduino operates at 5V and the Micro SD Card operates at 3.3V, a typical Micro SD Card Adapter or an SD Card Module basically consists of two important components. They are the 3.3V Voltage Regulator IC and a 5V to 3.3V Level Converter IC for the communication pins.

Pins of SD Card Module

Talking about pins, as I have mentioned that a Micro SD Card supports only SPI Communication, the SD Card Module has pins for SPI Communication. So, the pins on an SD Card Module are as follows.

- VCC – 5V
- GND – GND
- MOSI – Master OUT Slave IN (Input)
- MISO – Master IN Slave OUT (Output)
- SCK – SPI Clock (Input)
- CS – Chip Select (Input)

Arduino SD Card Module Interface

Now that we have seen a little bit about the SD Card Module, let us proceed with interfacing one with Arduino. First thing to remember is that the communication between Arduino and the SD Card Module is through SPI Interface.

Hence, you have to identify the SPI Pins on your Arduino Board. In case of Arduino UNO, the SPI Pins are as follows:

- MOSI – 11
- MISO – 12
- SCK – 13
- CS or SS – 10

If you are using Arduino Mega, then check for the SPI Pins before making the connection.

Coming to the Arduino SD Card Module Interface, I have designed two circuits for this project. In the first circuit, I have simply made the connection between the Arduino and the SD Card Module and extract the information of the card. This circuit can be considered as an Arduino SD Card Module Hook-up Guide.

In the second circuit, the magic of actual data logging happens. It is an extension to the first circuit with sensors connected to the Analog Pins of Arduino and the data from these sensors is captured on an event.

Components Required

Components mentioned here are combined for both the circuits.

- Arduino UNO
- Micro SD Card
- Micro SD Card Adapter or SD Card Module
- Push Button
- 3 x 10KΩ Potentiometers
- Connecting Wires

ARDUINO UNO (ATMEGA 328) MICROCONTROLLER

Arduino/Genuino Uno is a microcontroller board based on the ATmega328P (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

"Uno" means one in Italian and was chosen to mark the release of Arduino Software (IDE) 1.0. The Uno board and version 1.0 of Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer releases. The Uno board is the first in a series of USB Arduino boards, and the reference model for the Arduino platform; for an extensive list of current, past or outdated boards see the Arduino index of boards.

Arduino is an open source, computer hardware and software company, project, and user community that designs and manufactures microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical world. The project's products are distributed as open-source hardware and software, which are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL),[1] permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially in preassembled form, or as do-it-yourself kits.

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a

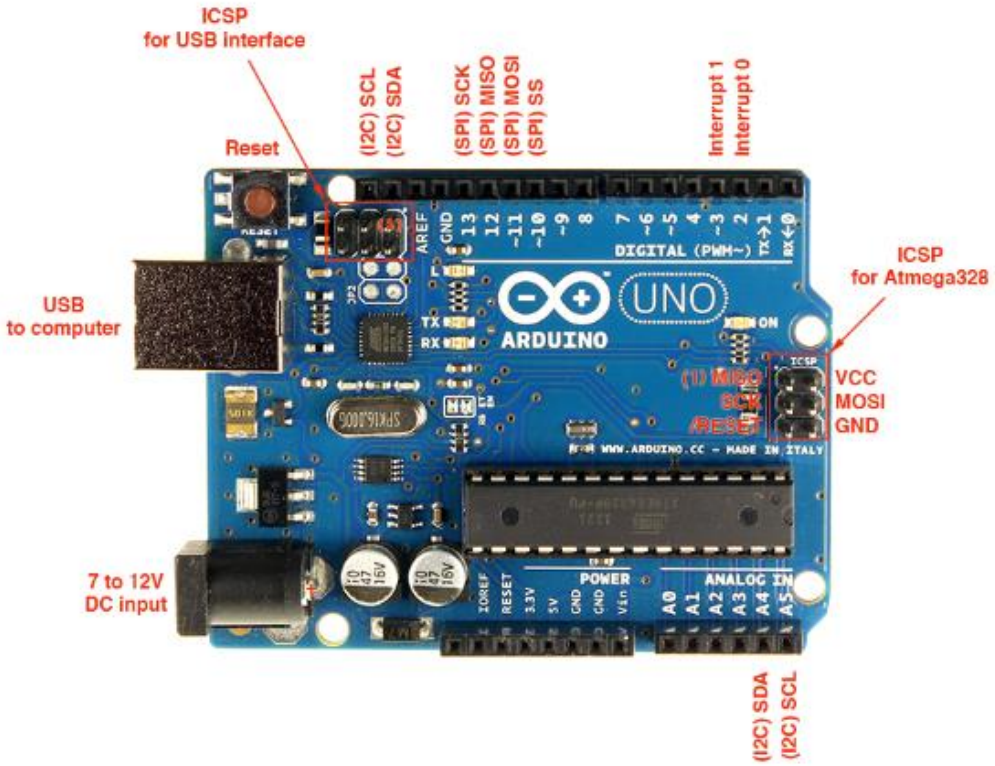
computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode Revision 3 of the board has the following new features: 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible with both the board that uses the AVR, which operates with 5V and with the Arduino Due that operates with 3.3V. The second one is a not connected pin, that is reserved for future purposes. Stronger RESET circuit. Atmega 16U2 replace the 8U2. "Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The microcontrollers are typically programmed using a dialect of features from the programming languages C and C++. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.

The Arduino project started in 2003 as a program for students at the Interaction Design Institute Ivrea in Ivrea, Italy,[2] aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats, and motion detectors.

HARDWARE

Arduino is open-source hardware. The hardware reference designs are distributed under a Creative Commons Attribution Share-Alike 2.5 license and are available on the Arduino website. Layout and production files for some versions of the hardware are also available. The source code for the IDE is released under the GNU General Public License, version 2.[8] Nevertheless, an official Bill of Materials of Arduino boards has never been released by Arduino staff.



SPECIFICATION

- Microcontroller ATmega328
- Operating Voltage 5V Input Voltage (recommended) 7-12V
- Input Voltage (limits) 6-20V
- Digital I/O Pins 14 (of which 6 provide PWM output)
- Analog Input Pins 6
- DC Current per I/O Pin 40 mA
- DC Current for 3.3V Pin 50 mA
- Flash Memory 32 KB (ATmega328) of which 0.5 KB used by bootloader SRAM 2 KB (ATmega328) EEPROM 1 KB (ATmega328) Clock Speed 16 MHz

Input and Output

Each of the 14 digital pins on the Arduino Uno can be used as an input or output, using pinMode(), digitalWrite(), and digitalRead() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms.
In addition, some pins have specialized functions:

Serial: pins 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.

External Interrupts: pins 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attachInterrupt() function for details.

PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the analogWrite() function.

SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication using the SPI library.

LED: 13. There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though is it possible to change the upper end of their range using the AREF pin and the analogReference() function. Additionally, some pins have specialized functionality:

TWI: A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.

There are a couple of other pins on the board:

AREF. Reference voltage for the analog inputs. Used with analogReference().

Reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

Digital Pins

In addition to the specific functions listed below, the digital pins on an Arduino board can be used for general purpose input and output via the pinMode(), digitalRead(), and digitalWrite() commands. Each pin has an internal pull-up resistor which can be turned on and off using digitalWrite() (w/ a value of HIGH or LOW, respectively) when the pin is configured as an input. The maximum current per pin is 40 mA.

Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. On the Arduino Diecimila, these pins are connected to the corresponding pins of the FTDI USB-to-TTL Serial chip. On the Arduino BT, they are connected to the corresponding pins of the WT11 Bluetooth module. On the Arduino Mini and LilyPad Arduino, they are intended for use with an external TTL serial module (e.g. the Mini-USB Adapter).

External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attachInterrupt() function for details.

PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the analogWrite() function. On boards with an ATmega8, PWM output is available only on pins 9, 10, and 11.

BT Reset: 7. (Arduino BT-only) Connected to the reset line of the bluetooth module.

SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.

LED: 13. On the Diecimila and LilyPad, there is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

Analog Pins

In addition to the specific functions listed below, the analog input pins support 10-bit analog-to-digital conversion (ADC) using the analogRead() function. Most of the analog inputs can also be used as digital pins: analog input 0 as digital pin 14 through analog input 5 as digital pin 19. Analog inputs 6 and 7 (present on the Mini and BT) cannot be used as digital pins.

I^2C: 4 (SDA) and 5 (SCL). Support I^2C (TWI) communication using the Wire library (documentation on the Wiring website).

Power Pins

VIN (sometimes labelled "9V"). The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin. Note that different boards accept different input voltages ranges, please see the documentation for your board. Also note that the LilyPad has no VIN pin and accepts only a regulated input.

5V. The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.

3V3. (Diecimila-only) A 3.3 volt supply generated by the on-board FTDI chip.

GND. Ground pins.

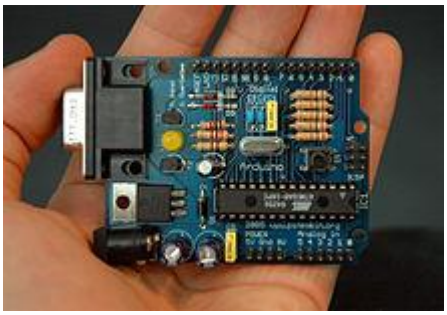
Other Pins

AREF. Reference voltage for the analog inputs. Used with analogReference().

Reset. (Diecimila-only) Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

Although the hardware and software designs are freely available under copyleft licenses, the developers have requested that the name Arduino be exclusive to the official product and not be used for derived works without permission. The official policy document on use of the Arduino name emphasizes that the project is open to

incorporating work by others into the official product. Several Arduino-compatible products commercially released have avoided the project name by using various names ending in arduino.



An early Arduino board[11] with an RS-232 serial interface (upper left) and an Atmel ATmega8 microcontroller chip (black, lower right); the 14 digital I/O pins are at the top, the 6 analog input pins at the lower right, and the power connector at the lower left.

Most Arduino boards consist of an Atmel 8-bit AVR microcontroller (ATmega8, ATmega168, ATmega328, ATmega1280, ATmega2560) with varying amounts of flash memory, pins, and features[12]. The 32-bit Arduino Due, based on the Atmel SAM3X8E was introduced in 2012[13]. The boards use single or double-row pins or female headers that facilitate connections for programming and incorporation into other circuits. These may connect with add-on modules termed shields. Multiple, and possibly stacked shields may be individually addressable via an I²C serial bus. Most boards include a 5 V linear regulator and a 16 MHz crystal oscillator or ceramic resonator. Some designs, such as the LilyPad, run at 8 MHz and dispense with the onboard voltage regulator due to specific form-factor restrictions.

Arduino microcontrollers are pre-programmed with a boot loader that simplifies uploading of programs to the on-chip flash memory. The default bootloader of the Aduino UNO is the optiboot bootloader.[14] Boards are loaded with program code via a serial connection to another computer. Some serial Arduino boards contain a level shifter circuit to

convert between RS-232 logic levels and transistor–transistor logic (TTL) level signals. Current Arduino boards are programmed via Universal Serial Bus (USB), implemented using USB-to-serial adapter chips such as the FTDI FT232. Some boards, such as later-model Uno boards, substitute the FTDI chip with a separate AVR chip containing USB-to-serial firmware, which is reprogrammable via its own ICSP header. Other variants, such as the Arduino Mini and the unofficial Boarduino, use a detachable USB-to-serial adapter board or cable, Bluetooth or other methods, when used with traditional microcontroller tools instead of the Arduino IDE, standard AVR in-system programming (ISP) programming is used.

The Arduino board exposes most of the microcontroller's I/O pins for use by other circuits. The Diecimila, Duemilanove, and current Uno[provide 14 digital I/O pins, six of which can produce pulse-width modulated signals, and six analog inputs, which can also be used as six digital I/O pins. These pins are on the top of the board, via female 0.1-inch (2.54 mm) headers. Several plug-in application shields are also commercially available. The Arduino Nano, and Arduino-compatible Bare Bones Board[15] and Boarduino[16] boards may provide male header pins on the underside of the board that can plug into solderless breadboards.

Many Arduino-compatible and Arduino-derived boards exist. Some are functionally equivalent to an Arduino and can be used interchangeably. Many enhance the basic Arduino by adding output drivers, often for use in school-level education, to simplify making buggies and small robots. Others are electrically equivalent but change the form factor, sometimes retaining compatibility with shields, sometimes not. Some variants use different processors, of varying compatibility.

TTC5200 - HIGH POWER AMPLIFIER TRANSISTOR

TTC5200 - NPN Transistor designed for general purpose power amplifier and switching applications. This is an original Toshiba transistor for high power audio amplifiers.

An audio power amplifier (or power amp) is an electronic amplifier that amplifies low-power electronic audio signals such as the signal from radio receiver or electric guitar pickup to a level that is high enough for driving loudspeakers or headphones. Audio power amplifiers are found in all manner of sound systems including sound reinforcement, public address and home audio systems and musical instrument amplifiers like guitar amplifiers. It is the final electronic stage in a typical audio playback chain before the signal is sent to the loudspeakers.

The preceding stages in such a chain are low power audio amplifiers which perform tasks like pre-amplification of the signal (this is particularly associated with record turntable signals, microphone signals and electric instrument signals from pickups, such as the electric guitar and electric bass), equalization (e.g., adjusting the bass and treble), tone controls, mixing different input signals or adding electronic effects such as reverb. The inputs can also be any number of audio sources like record players, CD players, digital audio players and cassette players. Most audio power amplifiers require these low-level inputs, which are line level.

While the input signal to an audio power amplifier, such as the signal from an electric guitar, may measure only a few hundred microwatts, its output may be a few watts for small consumer electronics devices, such as clock radios, tens or hundreds of watts for a home stereo system, several thousand watts for a nightclub's sound system or tens of thousands of watts for a large rock concert sound reinforcement system. While power amplifiers are available in standalone units, typically aimed at the hi-fi audiophile market (a niche market) of audio enthusiasts and sound reinforcement system professionals, most consumer

electronics audio products, such as clock radios, boomboxes and televisions have relatively small power amplifiers that are integrated into the product.

Filters and preamplifiers[edit]

Since modern digital devices, including CD and DVD players, radio receivers and tape decks already provide a "flat" signal at line level, the preamp is not needed other than as a volume control and source selector. One alternative to a separate preamp is to simply use passive volume and switching controls, sometimes integrated into a power amplifier to form an *integrated amplifier*.

Power output stages[edit]

The final stage of amplification, after preamplifiers, is the output stage, where the highest demands are placed on the transistors or tubes. For this reason, the design choices made around the output device (for single-ended output stages, such as in single-ended triode amplifiers) or devices (for push-pull output stages), such as the Class of operation of the output devices is often taken as the description of the whole power amplifier. For example, a Class B amplifier will probably have just the high power output devices operating cut off for half of each cycle, while the other devices (such as differential amplifier, voltage amplifier and possibly even driver transistors) operate in Class A. In a transformerless output stage, the devices are essentially in series with the power supply and output load (such as a loudspeaker), possibly via some large capacitor and/or small resistances.

Further developments[edit]

For some years following the introduction of solid state amplifiers, their perceived sound did not have the excellent audio quality of the best valve amplifiers (see valve audio amplifier). This led audiophiles to believe that "tube sound" or valve sound had an intrinsic quality due

to the vacuum tube technology itself. In 1970, Matti Otala published a paper on the origin of a previously unobserved form of distortion: transient intermodulation distortion (TIM),[7] later also called slew-induced distortion (SID) by others. TIM distortion was found to occur during very rapid increases in amplifier output voltage.

TIM did not appear at steady state sine tone measurements, helping to hide it from design engineers prior to 1970. Problems with TIM distortion stem from reduced open loop frequency response of solid state amplifiers. Further works of Otala and other authors found the solution for TIM distortion, including increasing slew rate, decreasing preamp frequency bandwidth, and the insertion of a lag compensation circuit in the input stage of the amplifier. In high quality modern amplifiers the open loop response is at least 20 kHz, canceling TIM distortion.

The next step in advanced design was the Baxandall Theorem, created by Peter Baxandall in England.[13] This theorem introduced the concept of comparing the ratio between the input distortion and the output distortion of an amplifier. This new idea helped audio design engineers to better evaluate the distortion processes within an amplifier.

APPLICATIONS

Important applications include public address systems, theatrical and concert sound reinforcement systems, and domestic systems such as a stereo or home-theatre system. Instrument amplifiers including guitar amplifiers and electric keyboard amplifiers also use audio power amplifiers. In some cases, the power amplifier for an instrument amplifier is integrated into a single amplifier "head" which contains a preamplifier, tone controls, and electronic effects. These components may be mounted in a wooden speaker cabinet to create a "combo amplifier". Musicians with unique performance needs and/or a need for very powerful amplification may create a custom setup with

separate rackmount preamplifiers, equalizers, and a power amplifier mounted in a 19" road case.

Power amplifiers are available in standalone units, which are used by hi-fi audio enthusiasts and designers of public address systems (PA systems) and sound reinforcement systems. A hi-fi user of power amplifiers may have a stereo power amplifier to drive left and right speakers and a single-channel (mono) power amplifier to drive a subwoofer. The number of power amplifiers used in a sound reinforcement setting depends on the size of the venue. A small coffeehouse may have a single power amp driving two PA speakers. A nightclub may have several power amps for the main speakers, one or more power amps for the monitor speakers (pointing towards the band) and an additional power amp for the subwoofer. A stadium concert may have a large number of power amps mounted in racks. Most consumer electronics sound products, such as TVs, boom boxes, home cinema sound systems, Casio and Yamaha electronic keyboards, "combo" guitar amps and car stereos have power amplifiers integrated inside the chassis of the main product.

INTERNET OF THINGS (IOT)

The **Internet of things** (**IoT**) is the network of physical devices, vehicles, home appliances and other items embedded with electronics, software, sensors, actuators, and connectivity which enables these objects to connect and exchange data. Each thing is uniquely identifiable through its embedded computing system but is able to inter-operate within the existing Internet infrastructure.

The figure of online capable devices increased 31% from 2016 to 8.4 billion in 2017. Experts estimate that the IoT will consist of about 30 billion objects by 2020. It is also estimated that the global market value of IoT will reach $7.1 trillion by 2020.

The IoT allows objects to be sensed or controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems, and resulting in improved efficiency, accuracy and economic benefit in addition to reduced human intervention. When IoT is augmented with sensors and actuators, the technology becomes an instance of the more general class of cyber-physical systems, which also encompasses technologies such as smart grids, virtual power plants, smart homes, intelligent transportation and smart cities.

"Things", in the IoT sense, can refer to a wide variety of devices such as heart monitoring implants, biochip transponders on farm animals, cameras streaming live feeds of wild animals in coastal waters, automobiles with built-in sensors, DNA analysis devices for environmental/food/pathogen monitoring, or field operation devices that assist firefighters in search and rescue operations. Legal scholars suggest regarding "things" as an "inextricable mixture of hardware, software, data and service".

These devices collect useful data with the help of various existing technologies and then autonomously flow the data between other devices

As of 2016, the vision of the Internet of things has evolved due to a convergence of multiple technologies, including ubiquitous wireless communication, real-time analytics, machine learning, commodity sensors, and embedded systems. This means that the traditional fields of embedded systems, wireless sensor networks, control systems, automation(including home and building automation), and others all contribute to enabling the Internet of things.[18]

The concept of a network of smart devices was discussed as early as 1982, with a modified Coke machine at Carnegie Mellon University becoming the first Internet-connected appliance, able to report its inventory and whether newly loaded drinks were cold. Mark Weiser's seminal 1991 paper on ubiquitous computing, "The Computer of the 21st Century", as well as academic venues such as UbiComp and PerCom produced the contemporary

vision of IoT. In 1994 Reza Raji described the concept in *IEEE Spectrum* as "[moving] small packets of data to a large set of nodes, so as to integrate and automate everything from home appliances to entire factories". Between 1993 and 1996 several companies proposed solutions like Microsoft's at Work or Novell's NEST. However, only in 1999 did the field start gathering momentum. Bill Joy envisioned Device to Device (D2D)communication as part of his "Six Webs" framework, presented at the World Economic Forum at Davos in 1999.

The concept of the Internet of things became popular in 1999, through the Auto-ID Center at MIT and related market-analysis publications. Radio-frequency identification (RFID) was seen by Kevin Ashton (one of the founders of the original Auto-ID Center) as a prerequisite for the Internet of things at that point. Ashton prefers the phrase "Internet *for*things." If all objects and people in daily life were equipped with identifiers, computers could manage and store them. Besides using RFID, the tagging of things may be achieved through such technologies as near field communication, barcodes, QR codes and digital watermarking.

In its original interpretation, one of the first consequences of implementing the Internet of things by equipping all objects in the world with minuscule identifying devices or machine-readable identifiers would be to transform daily life. For instance, instant and ceaseless inventory control would become ubiquitous. A person's ability to interact with objects could be altered remotely based on immediate or present needs, in accordance with existing end-user agreements. For example, such technology could grant motion-picture publishers much more control over end-user private devices by remotely enforcing copyright restrictions and digital rights management, so the ability of a customer who bought a Blu-ray disc to watch the movie could become dependent on the copyright holder's decision, similar to Circuit City's failed DIVX.

A significant transformation is to extend "things" from the data generated from devices to objects in the physical space. The thought-model for future interconnection environment was proposed in 2004. The model includes the notion of the ternary universe consists of the physical world, virtual world and mental world and a multi-level reference architecture with the nature and devices at the bottom level followed by the level of the Internet, sensor network, and mobile network, and intelligent human-machine communities at the top level, which supports geographically dispersed users to cooperatively accomplish tasks and solve problems by using the network to actively promote the flow of material, energy, techniques, information, knowledge, and services in this environment. This thought model envisioned the development trend of the Internet of things.

APPLICATIONS

- **Consumer application**
- **Smart Home**
- **Media**
- **Infrastructure Management**
- **Manufacturing**
- **Agriculture**
- **Energy management**
- **Environmental monitoring**
- **Building and home automation**
- **Metropolitan scale deployments**
- **Medical and healthcare**
- **Transportation**
- **ESP 8266- 12E NODE MCU (IOT MODULE)**

NodeMCU is an open source IoT platform. It includes firmware which runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module. The term "NodeMCU" by default refers to the firmware rather than the dev kits. The firmware uses the Lua scripting language. It is based on the eLua project, and built on the Espressif Non-OS SDK for ESP8266. It uses many open source projects, such as lua-cjson, and spiffs.

The *Internet of things* (IoT) is the network of everyday objects — physical things embedded with electronics, software, sensors, and connectivity enabling data exchange. Basically, a little networked computer is attached to a thing, allowing information exchange to and from that thing. Be it lightbulbs, toasters, refrigerators, flower pots, watches, fans, planes, trains, automobiles, or anything else around you, a little networked computer can be combined with it to accept input (especially object control) or to gather and generate informational output (typically object status or other sensory data). This means computers will be permeating everything around us — ubiquitous embedded computing devices, uniquely identifiable, interconnected across the Internet. Because of low-cost, networkable microcontroller modules, the Internet of things is really starting to take off.

NodeMCU was created shortly after the ESP8266 came out. On December 30, 2013, Espressif Systems began production of the ESP8266. The ESP8266 is a Wi-Fi SoC integrated with a Tensilica Xtensa LX106 core, widely used in IoT applications. NodeMCU started on 13 Oct 2014, when Hong committed the first file of nodemcu-firmware to GitHub. Two months later, the project expanded to include an open-hardware platform when developer Huang R committed the gerber file of an ESP8266 board, named devkit v0.9. Later that month, Tuan PM ported MQTT client library from Contiki to the ESP8266 SoC platform, and committed to NodeMCU project, then NodeMCU was able to support the MQTT IoT protocol, using Lua to access the MQTT broker. Another important update was made on 30 Jan 2015, when Devsaurus ported the u8glib to NodeMCU project, enabling NodeMCU to easily drive LCD, Screen, OLED, even VGA displays.

ESP8266 ARDUINO CORE

Arduino began developing new MCU boards based on non-AVR processors like the ARM/SAM MCU and used in the Arduino Due, they needed to modify the Arduino IDE so that it would be relatively easy to change the IDE to support alternate tool chains to allow Arduino C/C++ to be compiled down to these new processors. They did this with the

introduction of the Board Manager and the SAM Core. A "core" is the collection of software components required by the Board Manager and the Arduino IDE to compile an Arduino C/C++ source file down to the target MCU's machine language. Some creative ESP8266 enthusiasts have developed an Arduino core for the ESP8266 WiFi SoC that is available at the GitHub ESP8266 Core webpage. This is what is popularly called the "ESP8266 Core for the Arduino IDE" and it has become one of the leading software development platforms for the various ESP8266 based modules and development boards, including NodeMCUs.

The Button is a Wi-Fi connected push button designed by Peter R Jennings.[11] The Button is designed for single-purpose, internet-enabled functions. When the button is pressed, a connection is made to a web server which will perform the desired task. Applications include a doorbell or panic button.

Node USB

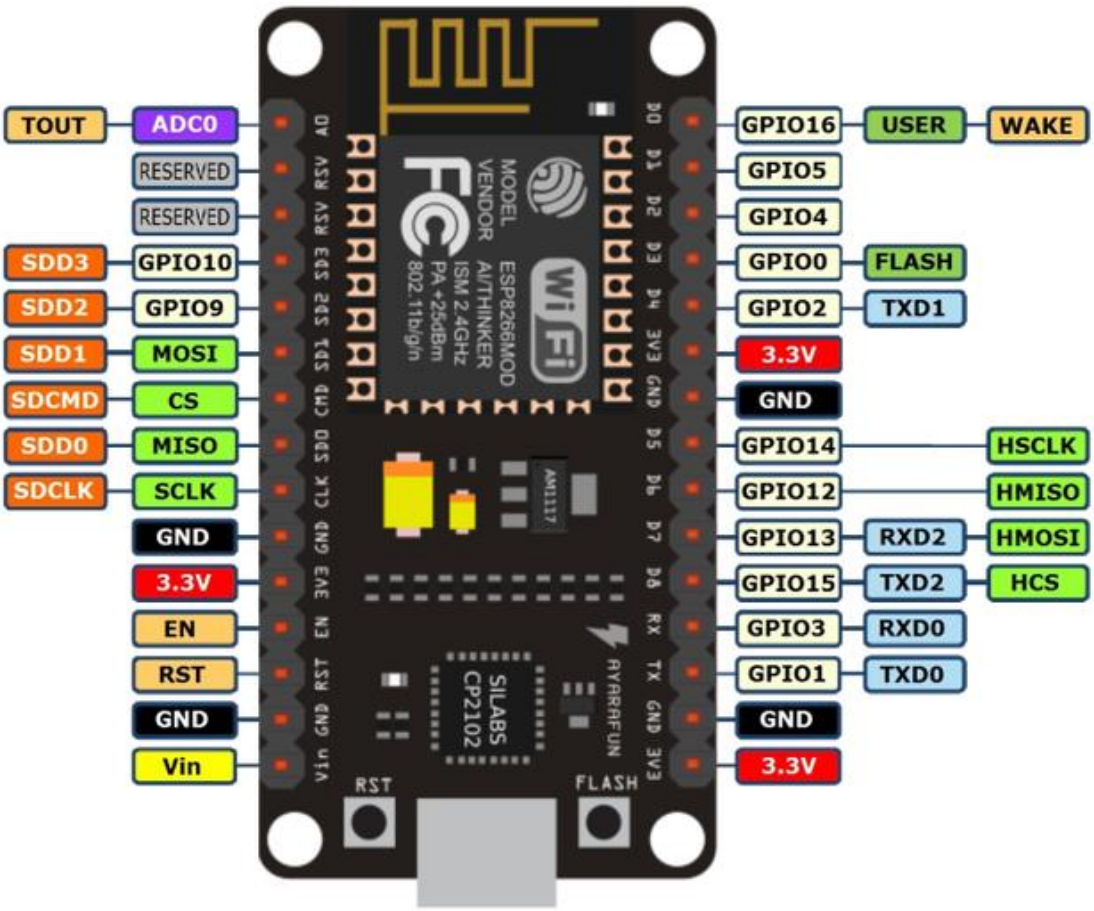
NodeUSB is an open IoT platform about the size of a standard USB stick. It was designed to leverage NodeMCU (Lua) for easy programming and has the extra feature of USBcapability. It is ideal for Plug-n-Play solutions, allowing easy prototyping for developers

NodeMCU provides access to the GPIO (General Purpose Input/Output) and for developing purposes below pin mapping table should be referenced.

IO index	ESP8266 pin	IO index	ESP8266 pin
0 [*]	GPIO16	7	GPIO13

1	GPIO5	8	GPIO15
2	GPIO4	9	GPIO3
3	GPIO0	10	GPIO1
4	GPIO2	11	GPIO9
5	GPIO14	12	GPIO10
6	GPIO12		

Node MCU is an open source IoT platform based on the ESP-12E module. The version 1.0 is the 5th design of NodeMCU devkit. This uses CP2102 as UART bridge, and can flash firmware automatically by using nodemcu-flasher. Also it has a voltage regulator to convert from 5v to 3.3v which is the required by the esp21e module.



	NodeMCU v1.0	**Arduino MKR1000**
Microcontroller	ESP-12E module, with Espressif ESP8266 32bits	ARM Cortex M0+ 32bits
Clock Speed	80/160MHz	48MHz
Board Power Supply	5V	5V

	NodeMCU v1.0	**Arduino MKR1000**
Circuit Operating Voltage	3.3V	3.3V
Flash Memory	4MB	256KB
SRAM	64KB	32KB
EEPROM	No	No
Digital I/O Pins	10	8
PWM Pins	10	12
Analog Input Pins	1 (ADC 10 bit)	7 (ADC 8/10/12 bit)
Analog Output Pins		1 (DAC 10 bit)
Connectivity	IEEE 802.11 b/g/n Wi-Fi	IEEE 802.11 b/g/n Wi-Fi
Antenna Type	PCB	PCB
Supported Battery	No	Li-Po single cell, 3.7V, 700mAh minimum

	NodeMCU v1.0	Arduino MKR1000
UART	1 (+ TX only on pin GPIO2)	1
SPI	2	1
I2C	1	1
LED builtin	D0 / GPIO 16	GPIO 6
Programming Languages	C++ / Python / Lua / Javascript	C++
Flashing	Locally / OTA	Locally / OTA

CHIP

The ESP8266 series, or family, of Wi-Fi chips is produced by Espressif Systems, a fabless semiconductor company operating out of Shanghai, China. The ESP8266 series presently includes the ESP8266EX and ESP8285 chips.

ESP8266EX (simply referred to as ESP8266) is a system-on-chip (SoC) which integrates a 32-bit Tensilica microcontroller, standard digital peripheral interfaces, antenna switches, RF balun, power amplifier, low noise receive amplifier, filters and power management modules into a small package. It provides capabilities for 2.4 GHz Wi-Fi (802.11 b/g/n, supporting WPA/WPA2), general-purpose input/output (16 GPIO), Inter-

Integrated Circuit (I^2C), analog-to-digital conversion (10-bit ADC), Serial Peripheral Interface (SPI), I^2S interfaces with DMA (sharing pins with GPIO), UART (on dedicated pins, plus a transmit-only UART can be enabled on GPIO2), and pulse-width modulation (PWM). The processor core, called "L106" by Espressif, is based on Tensilica's Diamond Standard 106Micro 32-bit processor controller core and runs at 80 MHz (or overclocked to 160 MHz). It has a 64 KiB boot ROM, 32 KiB instruction RAM, and 80 KiB user data RAM. (Also, 32 KiB instruction cache RAM and 16 KiB ETS system data RAM.) External flash memory can be accessed through SPI. The silicon chip itself is housed within a 5 mm × 5 mm Quad Flat No-Leads package with 33 connection pads — 8 pads along each side and one large thermal/ground pad in the center.

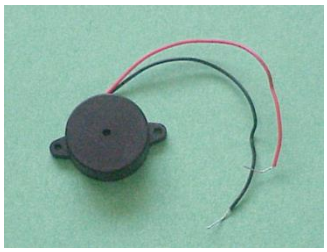
BUZZER

A buzzer or beeper is a signaling device, usually electronic, typically used in automobiles, household appliances such as a microwave oven, or game shows.

It most commonly consists of a number of switches or sensors connected to a control unit that determines if and which button was pushed or a preset time has lapsed, and usually illuminates a light on the appropriate button or control panel, and sounds a warning in the form of a continuous or intermittent buzzing or beeping sound. Initially this device was based on an electromechanical system which was identical to an electric bell without the metal gong (which makes the ringing noise). Often these units were anchored to a wall or ceiling and used the ceiling or wall as a sounding board. Another implementation with some AC-connected devices was to implement a circuit to make the AC current into a noise loud enough to drive a loudspeaker and hook this circuit up to a cheap 8-ohm speaker. Nowadays, it is more popular to use a ceramic-based piezoelectric sounder like a Sonalert which makes a high-pitched tone. Usually these were hooked up to "driver" circuits which varied the pitch of the sound or pulsed the sound on and off.

In game shows it is also known as a "lockout system," because when one person signals ("buzzes in"), all others are locked out from signalling. Several game shows have large buzzer buttons which are identified as "plungers".

The word "buzzer" comes from the rasping noise that buzzers made when they were electromechanical devices, operated from stepped-down AC line voltage at 50 or 60 cycles. Other sounds commonly used to indicate that a button has been pressed are a ring or a beep.



BUZZER

LCD DISPLAY

INTRODUCTION:

Liquid crystal cell displays (LCDs) are used in similar applications where LEDs are used. These applications are display of display of numeric and alphanumeric characters in dot matrix and segmental displays.

LCDs are of two types:

I. Dynamic scattering type
II. Field effect type

The construction of a dynamic scattering liquid crystal cell:

The liquid crystal material may be one of the several components, which exhibit optical properties of a crystal though they remain in liquid form. Liquid crystal is layered between glass sheets with transparent electrodes deposited on the inside faces.

When a potential is applied across the cell, charge carriers flowing through the liquid disrupt the molecular alignment and produce turbulence. When the liquid is not activated, it is transparent. When the liquid is activated the molecular turbulence causes light to be scattered in all directions and the cell appear to be bright.

This phenomenon is called dynamic scattering.

The construction of a field effect liquid crystal display is similar to that of the dynamic scattering type, with the exception that two thin polarizing optical filters are placed at the inside of each glass sheet. The liquid crystal material in the field effect cell is also of different type from employed in the dynamic scattering cell. The material used is twisted numeric type and actually twists the light passing through the cell when the latter is not energised.

A liquid crystal display (LCD) is an electronically-modulated optical device shaped into a thin, flat panel made up of any number of color or monochrome pixels filled with liquid crystals and arrayed in front of a light source (backlight) or reflector. It is often utilized in battery-powered electronic devices because it uses very small amounts of electric power. LCD has material, which continues the properties of both liquids and crystals. Rather than having a melting point, they have a temperature range within which the molecules are almost as mobile as they would be in a liquid, but are grouped together in an ordered from similar to a crystal.

LCD consists of two glass panels, with the liquid crystal materials sandwiched in between them. The inner surface of the glass plates is coated with transparent electrodes which define in between the electrodes and the crystal, which makes the liquid crystal

molecules to maintain a defined orientation angle. When a potential is applied across the cell, charge carriers flowing through the liquid will disrupt the molecular alignment and produce turbulence.

When the liquid is not activated, it is transparent. When the liquid is activated the molecular turbulence causes light to be scattered in all directions and the cell appears to be bright. Thus the required message is displayed. When the LCD is in the off state, the two polarizer's and the liquid crystal rotate the light rays, such that they come out of the LCD without any orientation, and hence the LCD appears transparent.

Working:

When sufficient voltage is applied to the electrodes the liquid crystal molecules would be aligned in a specific direction. The light rays passing through the LCD would be rotated by the polarizer, which would result in activating/highlighting the desired characters. The power supply should be of +5v, with maximum allowable transients of 10mv. To achieve a better/suitable contrast for the display the voltage (VL) at pin 3 should be adjusted properly. A module should not be removed from a live circuit.

The ground terminal of the power supply must be isolated properly so that voltage is induced in it. The module should be isolated properly so that stray voltages are not induced, which could cause a flicking display. LCD is lightweight with only a few, millimeters thickness since the LCD consumes less power, they are compatible with low power electronic circuits, and can be powered for long durations. LCD does not generate light and so light is needed to read the display. By using backlighting, reading is possible in the dark. LCDs have long life and a wide operating temperature range. Before LCD is used for displaying proper initialization should be done.

LCDs with a small number of segments, such as those used in digital watches and pocket calculators, have individual electrical contacts for each segment. An external

dedicated circuit supplies an electric charge to control each segment. This display structure is unwieldy for more than a few display elements. Small monochrome displays such as those found in personal organizers, or older laptop screens have a passive-matrix structure employing super-twisted nematic (STN) or double-layer STN (DSTN) technology—the latter of which addresses a color-shifting problem with the former—and color-STN (CSTN)— wherein color is added by using an internal filter. Each row or column of the display has a single electrical circuit.

The pixels are addressed one at a time by row and column addresses. This type of display is called passive-matrix addressed because the pixel must retain its state between refreshes without the benefit of a steady electrical charge. As the number of pixels (and, correspondingly, columns and rows) increases, this type of display becomes less feasible.

Very slow response times and poor contrast are typical of passive matrix addressed LCDs. High-resolution color displays such as modern LCD computer monitors and televisions use an active matrix structure. A matrix of thin-film transistors (TFTs) is added to the polarizing and color filters. Each pixel has its own dedicated transistor, allowing each column line to access one pixel. When a row line is activated, all of the column lines are connected to a row of pixels and the correct voltage is driven onto all of the column lines.

The row line is then deactivated and the next row line is activated. All of the row lines are activated in sequence during a refresh operation. Active-matrix addressed displays look "brighter" and "sharper" than passive-matrix addressed displays of the same size, and generally have quicker response times, producing much better images. A general purpose alphanumeric LCD, with two lines of 16 characters. So the type of LCD used in this project is16 characters * 2 lines with 5*7 dots with cursor, built in controller, +5v power supply, 1/16 duty cycle.



PIN DESCRIPTION FOR LCD:

PIN NO	SYMBOL	FUNCTION
1	Vss	Ground terminal of Module
2	Vdd	Supply terminal of Module, + 5v
3	Vo	Power supply for liquid crystal drive
4	RS	Register select RS=0…Instruction register RS=1…Data register
5	R/W	Read/Write R/W=1…Read R/W=0…Write

6	EN	Enable
7-14	DB0-DB7	Bi-directional Data Bus. Data Transfer is performed once ,thru DB0-DB7,incase of interface data length is 8-bits;and twice, thru DB4-DB7 in the case of interface data length is 4-bits.Upper four bits first then lower four bits.
15	LAMP-(L-)	LED or EL lamp power supply terminals
16	LAMP+(L+) (E2)	Enable

LCD PIN DESCRIPTIONS:

The function of each pins of LCD is described below **VCC, VSS and VEE** while v and v provide +5v and ground, respectively, v is used for controlling LCD contrast.

RS, register select

There are two very important registers inside the LCD. The RS pin is used for their selection as follows. If RS=0, the instruction code register is selected, allowing the user to send a command such as clear display, cursor at home,etc.if RS=1 the data register is selected, allowing the user to send data to be displayed on the LCD.

R/W, read/write

R/W input allows the user to write information to the LCD or read information from it. R/W=1 when reading; R/W=0 when writing.

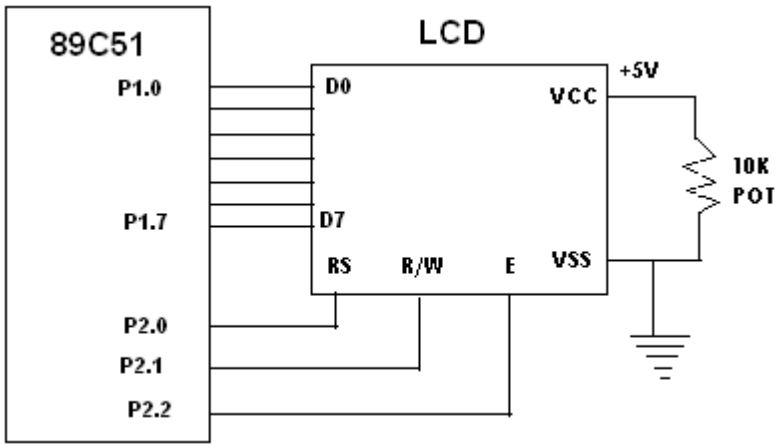
E, enable

The enable pin is used by the LCD to latch information presented on its data pins. When data is supplied to data pins, a high to low pulse must be applied to this pin in order for the LCD to latch in the data present at the data pins.

D0 - D7

The 8-bit data pins, D0 – D7, are used to send information to the LCD or read contents of the LCD'S internal registers. There are also instruction codes that can be sent to the LCD to clear the display or force the cursor to the home position or blink the cursor. RS=0 is used to check the busy flag bit to see if the LCD is ready to receive information. The busy flag is D7 and can be read when R/W=1 and RS=0, as follows: if R/W=1, RS=0.when D7=1, the LCD is busy taking care of internal operation and will not accept any new information, when D7=0, the LCD is ready to receive new information.

LCD INTERFACING WITH MICROCONTROLLER:



ADVANTAGES:

1. Consume much lesser energy (i.e. low power) when compared to LEDs.
2. Utilizes the light available outside and no generation of light.
3. Since very thin layer of liquid crystal is used, more suitable to act as display elements (in digital watches, pocket calculators, ect.)
4. Since reflectivity is highly sensitive to temperature, used as temperature measuring sensor.
5. Very cheap.

DISADVANTAGES:

1. Angle of viewing is very limited.
2. External light is a must for display.
3. Since not generating its own light and makes use of external light for display, contrast is poor.
4. Cannot be used under wide range of temperature.

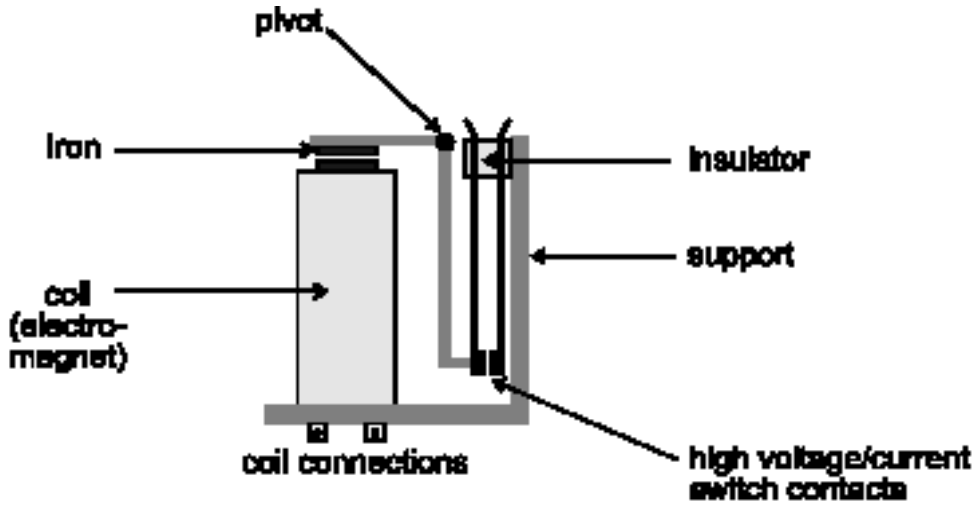
APPLICATIONS:

1. Watches
2. Fax & Copy machines & Calculators.

Relay Driver:

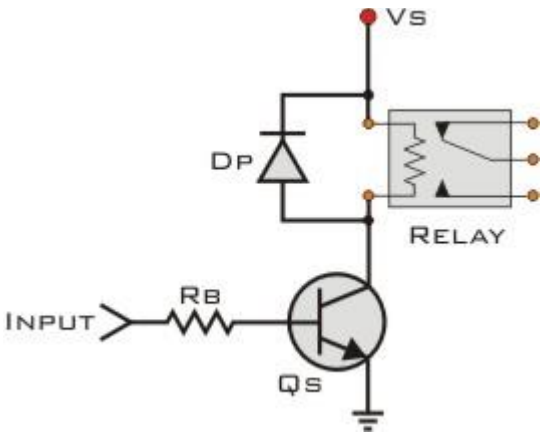
A relay is an electro-magnetic switch which is useful if you want to use a low voltage circuit to switch on and off a light bulb (or anything else) connected to the 220v mains supply.

The diagram below shows a typical relay (with “normally-open” contacts).



The current needed to operate the relay coil is more than can be supplied by most chips (op. amps etc), so a transistor is usually needed.

Relay Driver using single transistor:



This circuit will drive a relay coil from a low power output, usually from an IC like 555 or a TTL/CMOS. It is used to switch high loads or loads that needs AC current to operate. The relay will be actuated when the input of the circuit goes high. The protection diode Dp is used to protect the transistor from the reverse current generated from the coil of the relay during the switch off time. The values for Rb and Qs vary accordingly. The way to calculate them is:

First we calculate the load current:

IL = VS / RL

Then we calculate the transistor h_{FE}. It must be at least 5 times the load current IL divided by the maximum output current from the Input to the base of the transistor

$$h_{FE}(min) > 5 \text{ X } \frac{I_L}{I_{INPUT}}$$

Now you can choose the transistor Qs. You must select it according to it's Ic that must be greater than I_L and it's current gain h_{FE}.

Then you calculate the base resistor R_B, If the input is taken from a component (possible an IC) that uses the same power supply as the transistor (that is Vs), then the form is:

$$R_B = 0.2 \text{ X } R_L \text{ X } h_{FE}$$

Otherwise, if the component uses another power source (like V_{CC}) then the form is:

$$R_B = \frac{V_{CC} \text{ X } h_{FE}}{5 \text{ X } I_L}$$

The protective diode could be the 1N4001 or any general purpose diode.

An example:

The output from a 74LS series TTL IC is required to operate a relay with a 160 Ohm coil. The supply voltage is 12V for the transistor and 5V for the IC. The IC can supply a maximum current of 2mA.

$$I_L = Vs / R_L \Rightarrow I_L = 12 / 160 = 75mA$$

The transistor must have an h_{FE} greater than 5 X 75 / 2 => $\mathbf{h_{FE} > 187.5}$. So we choose a transistor with h_{FE} = 200 and IC = 100mA.

Now for the R_B resistor. Because the power supplies of transistor and I_C are different, we use the second formula:

$$\mathbf{R_B = \frac{V_{CC} \; X \; h_{FE}}{5 \; X \; I_L}}$$

With the use of some basic maths, this will produce $\mathbf{R_B = 2666.6\ Ohm}$. You choose the closest resistor possible to this value.

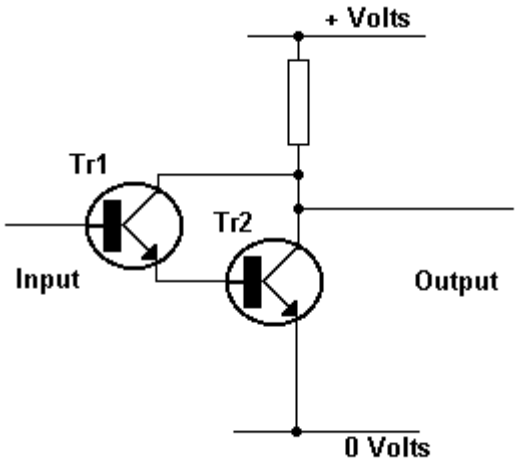
Darlington Pair

A Darlington pair is two transistors that act as a single transistor but with a much higher current gain.

The **Darlington transistor** (often called a **Darlington pair**) is a compound structure consisting of two bipolar transistors (either integrated or separated devices) connected in such a way that the current amplified by the first transistor is amplified further by the second one. This configuration gives a much higher current gain (written β, h_{fe}, or h_{FE}) than each transistor taken separately and, in the case of integrated devices, can take less space than two individual transistors because they can use a *shared* collector. Integrated Darlington pairs come packaged in transistor-like integrated circuit packages.

The Darlington configuration was invented by Bell Laboratories engineer Sidney Darlington in 1953. He patented the idea of having two or three transistors on a single chip

(and *sharing* a ***single*** *collector*), but not that of an arbitrary number (which might have covered all modern integrated circuits.



The circuit shows 'Darlington Pair' driver. The first transistor's emitter feeds into the second transistor's base and as a result the input signal is amplified by the time it reaches the output.

This circuit is used for amplifying currents, i.e., the amplified current from the first transistor is further amplified by the second transistor. Needless to say, this transistor combination exhibits a much higher current gain than if only one transistor was used.

The important point to remember is that the Darlington Pair is made up of two transistors and when they are arranged as shown in the circuit they are used to amplify weak signals.

Use of Darlington Pair

In some application the amount of input current available to switch on a transistor is very low. This may mean that a single transistor may not be able to pass sufficient current required by the load.

As stated earlier this equals the **input current** x **the gain of the transistor (hFE)**. If it is not be possible to increase the input current then we need to increase the gain of the transistor. This can be achieved by using a Darlington Pair.

A Darlington Pair acts as one transistor but with a current gain that equals:

Total current gain **(hFE total)** = current gain of transistor 1 **(hFE t1) x** current gain of transistor 2**(hFE t2)**

So for example if you had two transistors with a current gain **(hFE)** = 100:

(hFE total) = 100 **x** 100

(hFE total) = 10,000

You can see that this gives a vastly increased current gain when compared to a single transistor.

Therefore this will allow a very low input current to switch a much bigger load current.

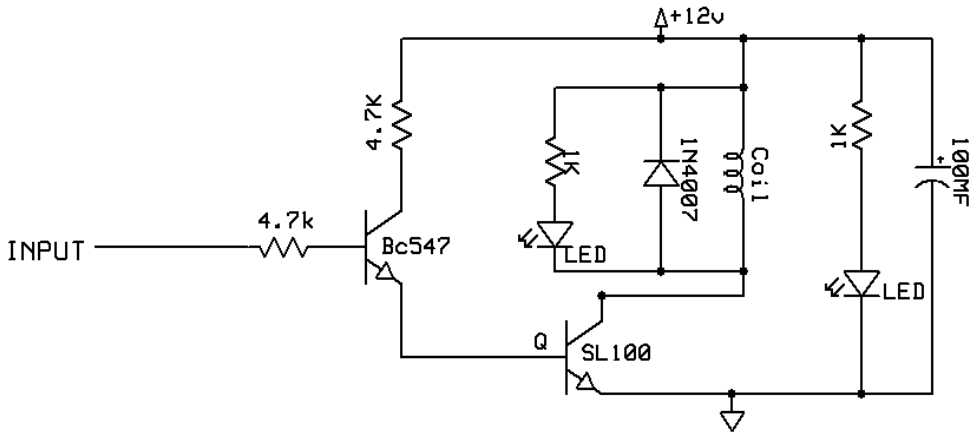
Base Activation Voltage

Normally to turn on a transistor the base input voltage of the transistor will need to be greater than 0.7V. As two transistors are used in a Darlington Pair this value is doubled.

Therefore the base voltage will need to be greater than 0.7V x 2 = 1.4V.

It is also worth noting that the voltage drop across collector and emitter pins of the Darlington Pair when the turn on will be around 0.9V Therefore if the supply voltage is 5V (as above) the voltage across the load will be will be around 4.1V (5V – 0.9V)

Circuit Diagram:



A Darlington pair behaves like a single transistor with a high current gain (approximately the product of the gains of the two transistors). In fact, integrated devices have three leads (B, C and E) broadly equivalent to those of a standard transistor.

A general relation between the compound current gain and the individual gains is given by:

If β_1 and β_2 are high enough (hundreds), this relation can be approximated with:

A typical modern device has a current gain of 1000 or more, so that only a small base current is needed to make the pair switch on. However, this high current gain comes with several drawbacks.

One drawback is an approximate doubling of base-emitter voltage. Since there are two junctions between the base and emitter of the Darlington transistor, the equivalent base-emitter voltage is the sum of both base-emitter voltages:

For silicon-based technology, where each V_{BEi} is about 0.65 V when the device is operating in the active or saturated region, the necessary base-emitter voltage of the pair is 1.3 V.

Another drawback of the Darlington pair is its increased saturation voltage. The output transistor is not allowed to saturate (i.e. its base-collector junction must remain reverse-biased) because its collector-emitter voltage is now equal to the sum of its own base-emitter voltage and the collector-emitter voltage of the first transistor, both positive quantities in normal operation. (In symbols, $V_{CE2} = V_{BE2} + V_{CE1}$, so $V_{C2} > V_{B2}$ always.) Thus the saturation voltage of a Darlington transistor is one V_{BE} (about 0.65 V in silicon) higher than a single transistor saturation voltage, which is typically 0.1 - 0.2 V in silicon. For equal collector currents, this drawback translates to an increase in the dissipated power for the Darlington transistor over a single transistor.

Another problem is a reduction in switching speed, because the first transistor cannot actively inhibit the base current of the second one, making the device slow to switch off. To alleviate this, the second transistor often has a resistor of a few hundred ohms connected between its base and emitter terminal. This resistor provides a low impedance discharge path for the charge accumulated on the base emitter junction, allowing a faster transistor turn-off.

The Darlington pair has more phase shift at high frequencies than a single transistor and hence can more easily become unstable with negative feedback.

A Darlington pair can be sensitive enough to respond to the current passed by skin contact even at safe voltages. Thus it can form the input stage of a touch-sensitive switch.

When current to the coil is switched off the magnetic field around the coil collapses very quickly. This can cause a phenomenon called "**Back emf**" where a very high voltage

Specification:

BC 547

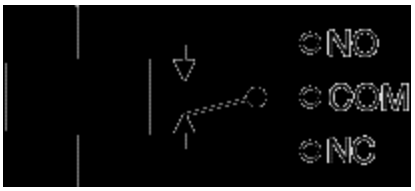
1. Current - 100 mA
2. Voltage - max. 65 V

SL 100

1. Current – 500mA
2. Voltage – max. 50V

RELAYS

A relay is an **electrically operated switch**. Current flowing through the coil of the relay creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off so relays have two switch positions and most have **double throw** (**changeover**) switch contacts as shown in the diagram.



Circuit symbol for a relay



Relays allow one circuit to switch a second circuit which can be completely separate from the first. For example a low voltage battery circuit can use a relay to switch a 230V AC mains circuit. There is no electrical connection inside the relay between the two circuits, the link is magnetic and mechanical.



Relays

The coil of a relay passes a relatively large current, typically 30mA for a 12V relay, but it can be as much as 100mA for

relays designed to operate from lower voltages. Most ICs (chips) cannot provide this current and a transistor is usually used to amplify the small IC current to the larger value required for the relay coil. The maximum output current for the popular 555 timer IC is 200mA so these devices can supply relay coils directly without amplification.

Relays are usuallly SPDT or DPDT but they can have many more sets of switch contacts, for example relays with 4 sets of changeover contacts are readily available.

The coil will be obvious and it may be connected either way round. Relay coils produce brief high voltage 'spikes' when they are switched off and this can destroy transistors and ICs in the circuit. To prevent damage you must connect a protection diode across the relay coil.

The relay's switch connections are usually labeled COM, NC and NO:

- **COM** = Common, always connect to this, it is the moving part of the switch.
- **NC** = Normally Closed, COM is connected to this when the relay coil is **off**.
- **NO** = Normally Open, COM is connected to this when the relay coil is **on**.
- Connect to COM and NO if you want the switched circuit to be **on when the relay coil is on**.
- Connect to COM and NC if you want the switched circuit to be **on when the relay coil is off**.

Choosing a relay

You need to consider several features when choosing a relay:

1. Physical size and pin arrangement

   If you are choosing a relay for an existing PCB you will need to ensure that its dimensions and pin arrangement are suitable. You should find this information in the supplier's catalogue.

2. Coil voltage

   The relay's coil voltage rating and resistance must suit the circuit powering the relay coil. Many relays have a coil rated for a 12V supply but 5V and 24V relays are also readily available. Some relays operate perfectly well with a supply voltage which is a little lower than their rated value.

3. Coil resistance

   The circuit must be able to supply the current required by the relay coil. You can use Ohm's law to calculate the current:

$$\text{Relay coil current} = \frac{\text{supply voltage}}{\text{coil resistance}}$$

   For example: A 12V supply relay with a coil resistance of 400 Ohm passes a current of 30mA. This is OK for a 555 timer IC (maximum output current 200mA), but it is too much for most ICs and they will require a transistor to amplify the current.

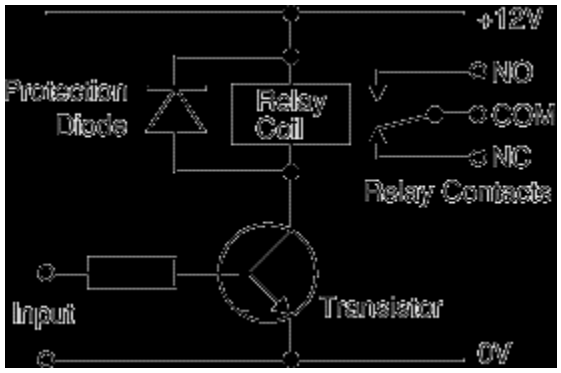
4. Switch ratings(voltage and current)

   The relay's switch contacts must be suitable for the circuit they are to control. You will need to check the voltage and current ratings. Note that the voltage rating is usually higher for AC, for example: "5A at 24V DC or 125V AC".

5. Switch contact arrangement(SPDT,DPDT etc)

Most relays are SPDT or DPDT which are often described as "single pole changeover" (SPCO) or "double pole changeover" (DPCO).

Protection diodes for relays



Transistors and ICs must be protected from the brief high voltage produced when a relay coil is switched off. The diagram shows how a signal diode (eg 1N4148) is connected 'backwards' across the relay coil to provide this protection.

Current flowing through a relay coil creates a magnetic field which collapses suddenly when the current is switched off. The sudden collapse of the magnetic field induces a brief high voltage across the relay coil which is very likely to damage transistors and ICs. The protection diode allows the induced voltage to drive a brief current through the coil (and diode) so the magnetic field dies away quickly rather than instantly. This prevents the induced voltage becoming high enough to cause damage to transistors and ICs.

Relays and transistors compared

Like relays, transistors can be used as an electrically operated switch. For switching small DC currents (< 1A) at low voltage they are usually a better choice than a relay. However, transistors cannot switch AC (such as mains electricity) and in simple circuits they are not

usually a good choice for switching large currents (> 5A). In these cases a relay will be needed, but note that a low power transistor may still be needed to switch the current for the relay's coil! The main advantages and disadvantages of relays are listed below:

Advantages of relays:

- Relays can switch **AC and DC**, transistors can only switch DC.
- Relays can switch **higher voltages** than standard transistors.
- Relays are often a better choice for switching **large currents** (> 5A).
- Relays can switch **many contacts** at once.

Disadvantages of relays:

- Relays are **bulkier** than transistors for switching small currents.
- Relays **cannot switch rapidly** (except reed relays), transistors can switch many times per second.
- Relays **use more power** due to the current flowing through their coil.
- Relays **require more current than many ICs can provide**, so a low power transistor may be needed to switch the current for the relay's coil.

CHAPTER – 5

SOFTWARE DESCRIPTION

SKETCH

In the getting started guide (Windows, Mac OS X, Linux), you uploaded a sketch that blinks an LED. In this tutorial, you'll learn how each part of that sketch works.

A *sketch* is the name that Arduino uses for a program. It's the unit of code that is uploaded to and run on an Arduino board.

Arduino is an open source, computer hardware and software company, project, and user community that designs and manufactures microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical world. The project's products are distributed as open-source hardware and software, which are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL),[1] permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially in preassembled form, or as do-it-yourself kits.

Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (*shields*) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The microcontrollers are

typically programmed using a dialect of features from the programming languages C and C++. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) based on the Processing language project.

The Arduino project started in 2003 as a program for students at the Interaction Design Institute Ivrea in Ivrea, Italy,[2] aiming to provide a low-cost and easy way for novices and professionals to create devices that interact with their environment using sensors and actuators. Common examples of such devices intended for beginner hobbyists include simple robots, thermostats, and motion detectors.

ARDUINO UNO

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new

needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

- **Inexpensive** - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than $50
- **Cross-platform** - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- **Simple, clear programming environment** - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- **Open source and extensible software** - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can

make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.

- **Open source and extensible hardware** - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

Variables

A *variable* is a place for storing a piece of data. It has a name, a type, and a value. For example, the line from the Blink sketch above declares a variable with the name ledPin, the type int, and an initial value of 13. It's being used to indicate which Arduino pin the LED is connected to. Every time the name ledPin appears in the code, its value will be retrieved. In this case, the person writing the program could have chosen not to bother creating the ledPin variable and instead have simply written 13 everywhere they needed to specify a pin number. The advantage of using a variable is that it's easier to move the LED to a different pin: you only need to edit the one line that assigns the initial value to the variable.

Functions

A *function* (otherwise known as a *procedure* or *sub-routine*) is a named piece of code that can be used from elsewhere in a sketch. For example, here's the definition of the setup() function from the Blink example:

```
void setup()
{
```

```
    pinMode(ledPin, OUTPUT);
}
```

The first line provides information about the function, like its name, "setup". The text before and after the name specify its return type and parameters: these will be explained later. The code between the { and } is called the *body* of the function: what the function does.

pinMode(), digitalWrite(), and delay()

The pinMode() function configures a pin as either an input or an output. To use it, you pass it the number of the pin to configure and the constant INPUT or OUTPUT. When configured as an input, a pin can detect the state of a sensor like a pushbutton; As an output, it can drive an actuator like an LED.

The digitalWrite() functions outputs a value on a pin.

For example, the line:

```
digitalWrite(ledPin, HIGH);
```

The delay() causes the Arduino to wait for the specified number of milliseconds before continuing on to the next line. There are 1000 milliseconds in a second, so the line:

```
delay(1000);
```

setup() and loop()

There are two special functions that are a part of every Arduino sketch: setup() and loop(). The setup() is called once, when the sketch starts. It's a good

place to do setup tasks like setting pin modes or initializing libraries. The loop() function is called over and over and is heart of most sketches. You need to include both functions in your sketch, even if you don't need them for anything.

Everything between the /* and */ is ignored by the Arduino when it runs the sketch (the * at the start of each line is only there to make the comment look pretty, and isn't required). It's there for people reading the code: to explain what the program does, how it works, or why it's written the way it is. It's a good practice to comment your sketches, and to keep the comments up-to-date when you modify the code. This helps other people to learn from or modify your code.

CHAPTER – 6

ADVANTAGES & APPLICATION

- Preserve perishable foods, finished products, unfinished material and process chemicals
- Monitor the physical condition of the inventory stock to prevent from damages
- Automate environment control – temperature and flame levels and PIR sensor
- Maintain a high-quality service standard for product storage and regulatory compliance with remote monitoring
- Receive notifications whenever environment parameters exceed the threshold
- As part of their research, the students tested a number of different frequencies. They found that high-pitched tones were ineffective on flames. Rather, low-frequency bass sounds between 30 and 60Hz were the most effective.

CHAPTER – 7

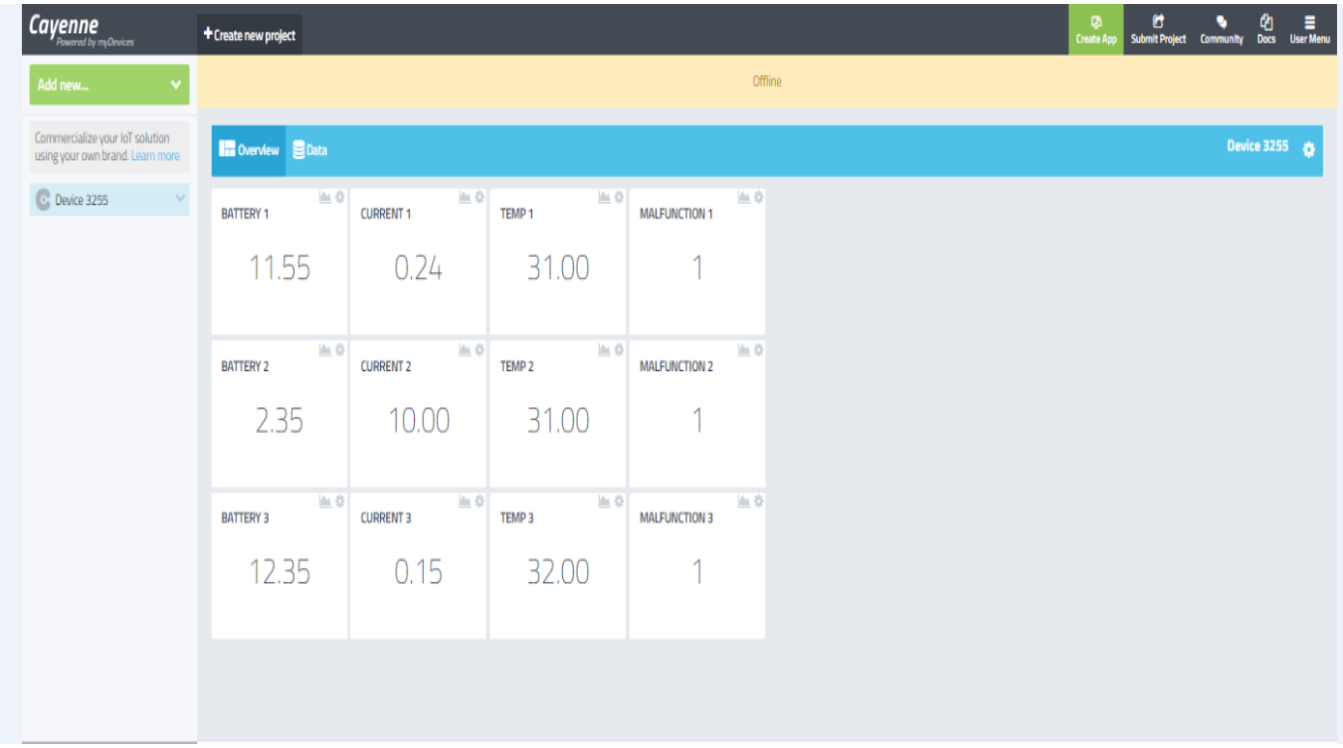
RESULTS AND CONCLUSION

Forest fires are on the rise in the United States of America, and all over the globe. Most of the fires are manmade. Technology can help prevent some of these fires. In this paper, IOT based forest fire detection was implemented using the Arduino Uno and acoustic wave based fire extinguisher system. So when the temperature and gas level is increased it will display on the LCD and also gives message to authorities by using IoT

module. Same process is implemented for the fire detection, when the flame gets increased the speaker will get activated and intimation will be conveyed to the authorities. So by using this technic we can protect the forests and we can save wild animals. In this paper, we investigated the performance of using acoustic wave based fire extinguisher system to detect and distinguish sounds.

SENDING DATA TO CLOUD

Internet of Things is a very fast growing technology. Integrating the warehouse fire detection system with IoT enables us to view the data obtained from various sensor and we can monitor anywhere with the help of our Mobile phone or it can be saved in Cloud and retrieved anytime for analysis. This feature is enabled by using a ESP 8266 node MCU module, which will collect the data from the controller and display it in cloud via any of the available bearer services. The IoT module can be checked by sending the MQTT protocol to it. By fixing appropriate delays, the request and response message from the IoT is fed to the Microcontroller and data is received from Microcontroller and sent to the webpage. The values are posted in the website server, which is specified in the bearer services of the IoT module. Once the connection with the bearer is established, the data can be transferred to the website and then monitored online. We have to provide the username, client ID, password of the website in which we need to post while configuring the ESP module. The IoT module is powered by the 5V.



Displaying Forest Fire Detection parameters on Webpages through IoT

Many organizations are therefore investing in IoT-enabled warehouses for better automated control systems (ACS) and warehouse management systems (WMS) to improve their operational efficiency by reducing costs. Our system providers have to handle thousands of environmentally sensitive items in the cold chain environment, their customers also have various handling requirements for storage conditions and cargo monitoring. The functional product quality is a primary concern for customers in selecting logistics service providers. In order to provide such services, cargo monitoring and effective storage guidance should be formulated so as to maintain a competitive edge in the market

CONCLUSION

This system is helpful to monitor the various parameters of warehouse and also it will inform to the customer by uploading the data on cloud computing server using IoT. The

recent improved processing capabilities of smart devices have shown promising results in surveillance systems for identification of different abnormal events i.e., fire, accidents, and other emergencies. Fire is one of the dangerous events which can result in great losses if it is not controlled on time.

CHAPTER – 8

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SMART AND SECURE WAREHOUSE MANAGEMENT SYSTEM USING IOT

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INTRODUCTION

- Warehouses are used by producers, dealers, traders, wholesalers, distributors, customs, etc

- To mitigate the manual labour work and to make the work easier.

- Using internet of things we can control system as well as we can access system remotely using IoT.

ABSTRACT

- We introduce a smart warehouse monitoring system, which is done with the help of current technology(IoT).
- The network of sensors includes humidity, temperature, fire, smoke sensors.
- The acoustic wave based fire extinguisher.
- We use Humidity sensor to measure the air moisture content of the Warehouse environment .

EXISTING SYSTEM

A.WMS: In the last decades, more advanced technologies used in storehouse management systems

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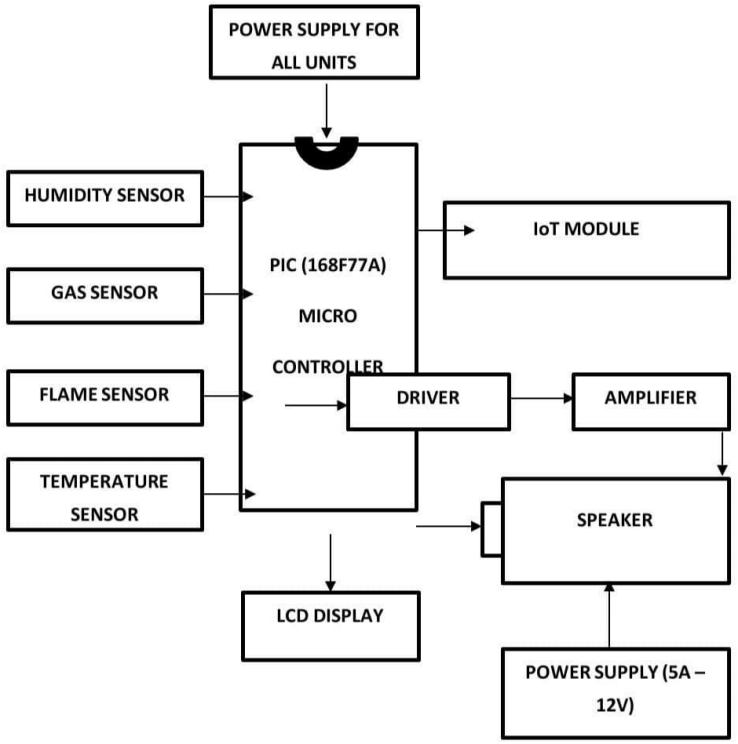
B. Data entry and Paperwork: Data entry and paper work has reduced the time working with the spread sheets and ledger maintenance of the management system.

C. Task Interleaving: It becomes more powerful tools so that system guidance will be extended to all the activities.

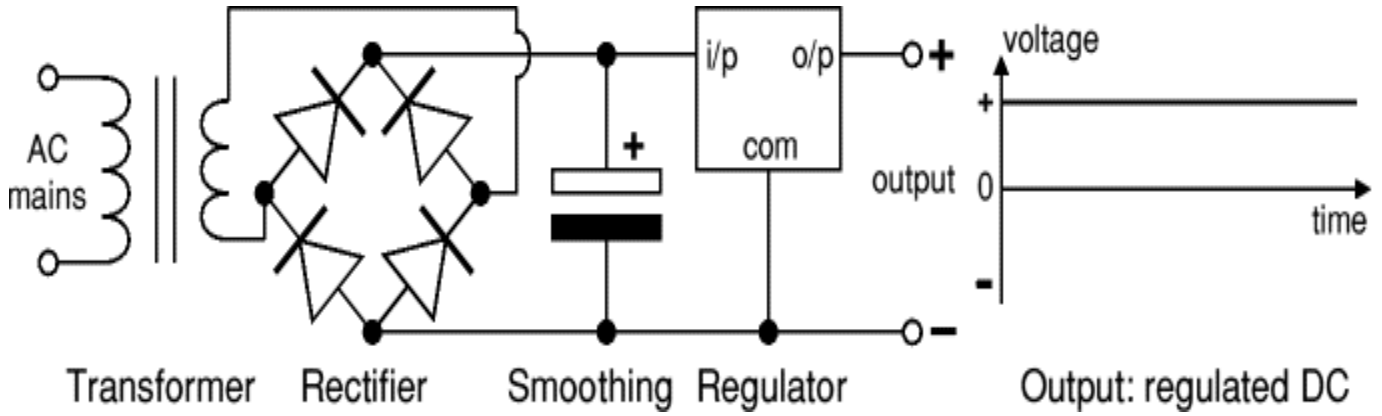
PROPOSED SYSTEM

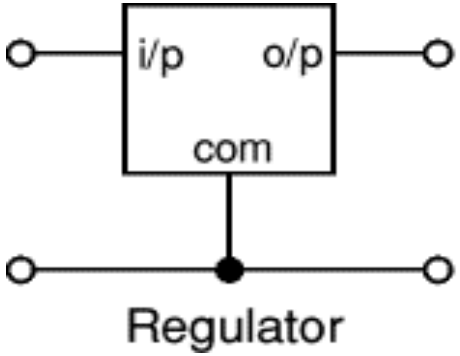
- The Gas sensor is used to detect If any gas is formed inside the Warehouse. Temperature and flame sensor is used to detect any fire.
- if fire is detected and then acoustic waves are generated to extinguish the fire
- Once we predict the fire or smoke by deep learning method, we will send a data to the ARDUINO UNO micro-controller from the arduino uno and various sensors.
- With the micro-controller, a SD card is connected and which is storing the sound files.

PROPOSED BLOCK DIAGRAM



POWER SUPPLY CIRCUIT DIAGRAM





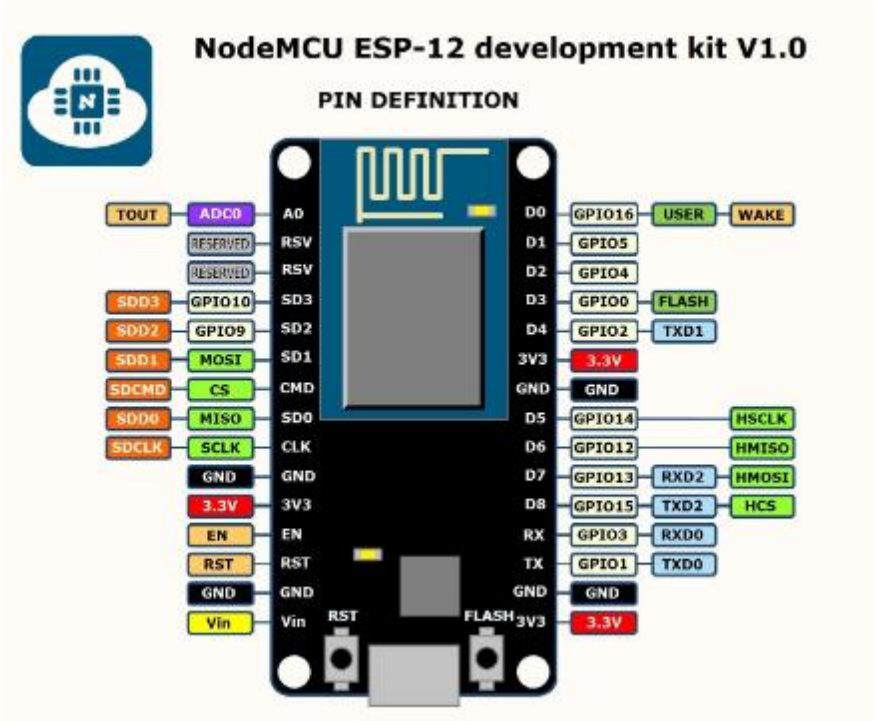
POWER SUPPLY

- The operation of power supply circuits built using filters, rectifiers and then voltage regulators.
- Starting with an AC voltage, a steady DC voltage is obtained by rectifying the AC voltage, then filtering to a DC level, and finally regulating to obtain a desired fixed DC voltage.
- The regulation is usually obtained from an IC voltage regulator unit, which remain the same if the input DC voltage varies or the output load connected to DC voltage changes.
- AC input ⟶ step down transformer ⟶ rectifier ⟶ filter ⟶ regulator
- A diode rectifier that provides a full-wave rectified voltage that is initially filtered by a simple capacitor filter to produce a DC voltage.
- A regulated circuit can use this DC inputs to provide a DC voltage that not only has much less ripple voltage but also remains the same DC value even if the input DC voltage varies somewhat or the load connected to the output DC voltage changes this voltage regulation is usually obtained using one of a number of popular voltage regulation IC unit.

ARDUINO UNO

- Arduino UNO is ATmega328 based microcontroller board.
- It is one of the most popular prototyping boards.
- The board comes with built-in arduino boot loader.
- It has 14 GPIO pins, 6 PWM pins, 6 Analog inputs and on board UART, SPI and TWI interfaces, an on-board resonator, a reset button, and holes for mounting pin headers.
- While programming the board, it can be connected to the PC using USB port and the board can runs on USB power.
- The Arduino UNO has 32 Kb Flash memory, 1 Kb EEPROM and 2 Kb SRAM.

ESP MODULE DEVELOPMENT KIT V1.0



IOT MODULE(ESP 8266 – 12E NODE MCU)

- **Node MCU** is an open source IoT platform.
- It includes firmware which runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module.
- The term "Node MCU" by default refers to the firmware rather than the dev kits.
- The firmware uses the Lua scripting language.
- It is based on the eLua project, and built on the Espressif Non-OS SDK for ESP8266.

FIRE EXTINGUISHER USING SONIC WAVES

- Sound wave could be one of the potential alternatives as fire extinguishers.

- The low frequency acoustic waves spilled from a speaker tend to extinguish the flames.



CONCLUSIONS

- By execution and implementation of the proposed technology the inventory management for the next generation smart storehouse management is proposed with Iot enabled sensor technology.

- The feature of this method or model is that it having more advance technology with enormous features.

- it is a future ready real time tracking system with cloud computing and support of the administration

ADVANTAGES

- Preserve perishable foods, finished products, unfinished material and process chemicals
- Monitor the physical condition of the inventory stock to prevent from damages
- Automate environment control – temperature and flame levels and PIR sensor
- Receive notifications whenever environment parameters exceed the threshold

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THANK YOU

Arduino Code - Team Project Evidence

This code is included as evidence of the project implementation. Vishwa has stated that another teammate was mainly responsible for the Arduino programming/configuration, so this should not be presented as code personally written by Vishwa.

```
#include "SD.h"
#define SD_ChipSelectPin 10
#include "TMRpcm.h"
#include "SPI.h"

#include<LiquidCrystal.h>
LiquidCrystal LCD(2,3,4,5,6,7);

int A=0,B=0,C=1;
TMRpcm tmrpcm;

void sdcard();

void setup() {
  pinMode(8,INPUT);
  pinMode(A0,INPUT);
  pinMode(A2,OUTPUT);

  LCD.begin(16,2);
  LCD.setCursor(0,0);
  LCD.print("   WELCOME   ");

  tmrpcm.speakerPin=9;
  Serial.begin(9600);

  if(!SD.begin(SD_ChipSelectPin)) {
    LCD.setCursor(0,0);
    LCD.print("SD fail");
    return;
  }

  delay(2000);
  LCD.clear();
}

void loop() {
  int a,b,c,d,e;

  a=digitalRead(8);
  b=analogRead(A0)/2.045;
  b=b-20;

  LCD.setCursor(0,0);
  LCD.print("T=");
  Serial.print('T');
  LCD.setCursor(2,0);
  LCD.print(b);
  Serial.print(b);

  if(a==1) {
    LCD.setCursor(0,1);
    LCD.print("no fire detected");
    Serial.print('F');
    delay(600);
    Serial.print('0');
    delay(1000);
  }

  if(a==0) {
    LCD.setCursor(0,1);
    LCD.print("  fire detected   ");
    Serial.print('F');
    delay(600);
    Serial.print('1');
    digitalWrite(A2,HIGH);
    delay(2000);
    digitalWrite(A2,LOW);
    B=0;
    A=1;
    sdcard();
    delay(2000);
  }
  LCD.clear();
```

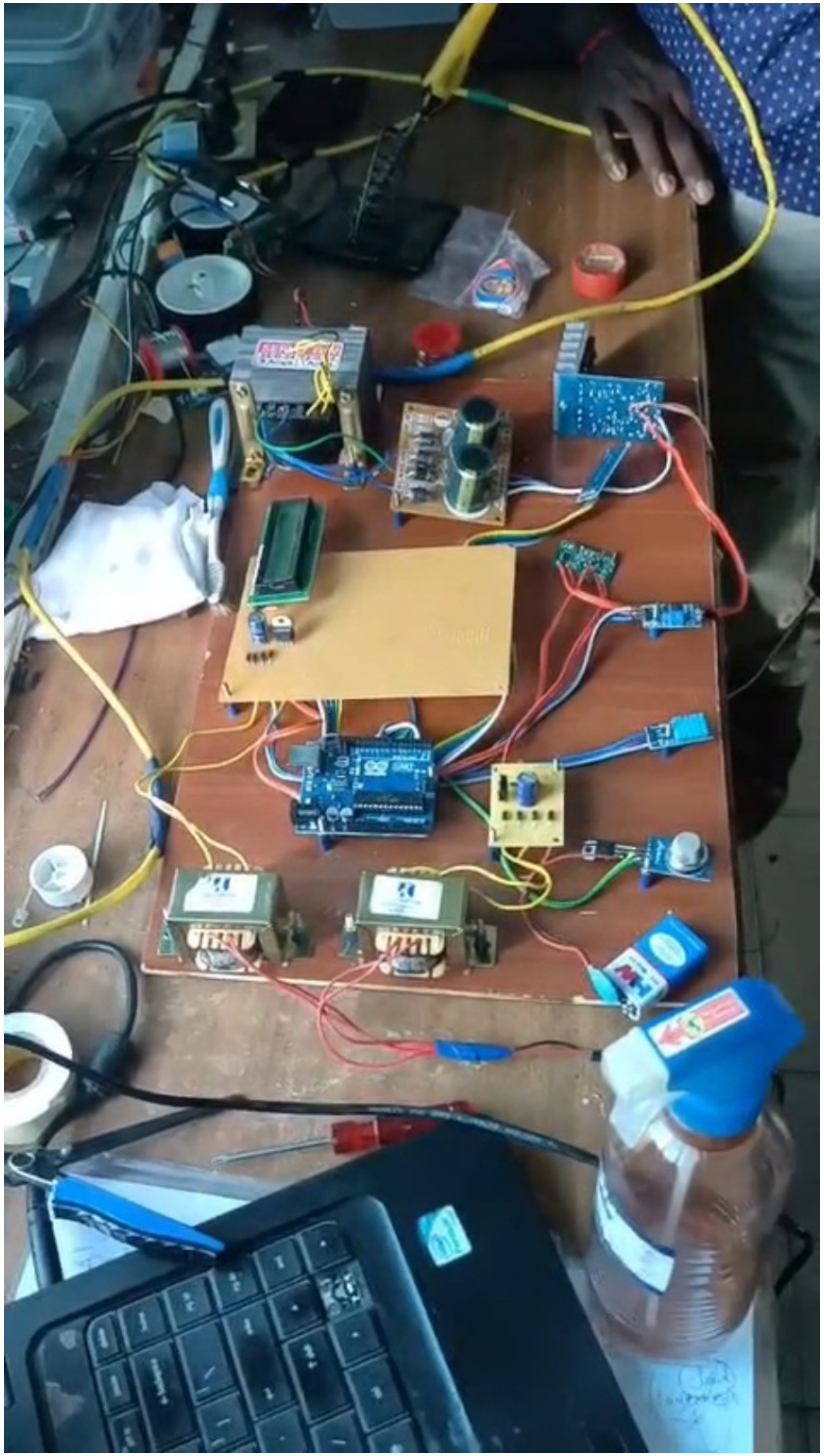
```
}

void sdcard() {
  C=0;
  {
    B=0;
    A=1;
  }

  if(A==1) {
    A=0;
    LCD.setCursor(0,0);
    LCD.print("   speaker on  ");
    tmrpcm.setVolume(6);
    tmrpcm.play("test.wav");
    delay(10000);
    tmrpcm.stopPlayback();
    C=1;
    LCD.clear();
  }

  if(B==1) {
    B=0;
    C=1;
    tmrpcm.stopPlayback();
  }
}
```

Prototype / testing photograph 1



Prototype / testing photograph 2

